

Sangbunhak

Sangbun Vector Theory

Concepts · Detailed Calculations · Graphs

Definitive Text with Integrated Calculations

From A1 to A12

Concepts and construction definitions: pages 2-12

Detailed calculations: pages 13-31

Graphs: pages 32-56

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Overview

Sangbunhak is a geometric and vector methodology that analyzes, step by step, geometric structures defined at a point and the vector relationships generated around that point.

It explores how tangents, normals, projections, rotations, and vector sums combine into a single vector structure, using intuitive geometric interpretation.

A Sangbun vector has a dual structure: a linear geometric component from tangents and projections, and a rotational geometric component from circles and rotations.

Assumptions

Vectors are calculated at a point on a line or curve. The constructions assume a nondegenerate general position. Special cases with undefined intersections require separate rules.

Z1 calculation: basic setup

Slope: a ; origin: $O(0, 0)$

Tangent: $y = ax + b$; point: $A(x, ax + b)$

Basic projections and intersections

Z1: Construction Points

A1: foot of the perpendicular from O to the tangent.

A2: foot of the perpendicular from O to the normal at A.

A3a: intersection of the normal at A with the x-axis.

$A3a' = (\text{x-coordinate of } A3a, \text{y-coordinate of } A1).$

A3b: intersection of the normal at A with the y-axis.

$A3b' = (\text{x-coordinate of } A1, \text{y-coordinate of } A3b).$

A3c: intersection of the vertical line through A3a with the tangent.

A3d: intersection of the horizontal line through A3b with the tangent.

A4a and A4b: intersections of two lines

A4a: intersection of OA3c and AA3a'.

A4b: intersection of OA3d and AA3b'.

Construction of A6 (vector sum)

Line (1) passes through A2 and is perpendicular to A1A2.

A5a: intersection of line (1) with the extended line A1A4a.

A5b: intersection of line (1) with the extended line A1A4b.

Vector $A1A6 = \text{vector } A1A5a + \text{vector } A1A5b.$

Circle (1) and principal points

Circle (1): center A1, radius $|A1A|$.

A7: the intersection with circle (1) beyond A1, along the ray from O through A1.

A8: the intersection other than A of circle (1) and the line through A perpendicular to vector OA.

A9, A10', and A11

A9 and A9a

From the midpoint of A7A8, move perpendicularly by $|A7A8|/2$ in each direction. A9 is the candidate inside circle (1); A9a is the candidate outside it.

Branch parameter

$$@ = |OA1| / |AA1|$$

Unified definition of A10'

A10' is the intersection of the line through A10 perpendicular to AA10 with the extended line from A toward the reference point. The reference point is A9 for $@ < 1$ and A7 for $@ > 1$.

Unified definition of A11

A, A10, and A10' form a triangle with its right angle at A10. A and A10' are the endpoints of the hypotenuse. The foot of the perpendicular from A10 to that hypotenuse is A9 for $@ < 1$ and A7 for $@ > 1$.

A11 is obtained by rotating the right-angle vertex A10 through 90° about A. Take as reference the rotation from the direction AO toward AA10. Use that same direction for $@ < 1$ and $@ = 1$, and the opposite direction for $@ > 1$.

$$|AA10| = |AA11|$$

A10' is not merely an auxiliary point arising during construction: it is introduced to accommodate $90^\circ + \theta$ at A after this rotation. Here θ is the angle at A between AA9 (AA7 when $@ > 1$) and AA10.

The Three Branches

Branch 1: $@ < 1$

Circle (2): center A_8 , radius $|A_8A_9|$.

A_{10} : the intersection other than A_9 of circle (2) and the line through A_9 perpendicular to AA_9 .

A_{10}' and A_{11} : use their unified definitions.

Branch 2: $@ > 1$

Circle (2): center A_9 , radius $|A_9A_7|$.

A_{10} : the intersection other than A_7 of circle (2) and the line through A_7 perpendicular to AA_7 .

A_{10}' and A_{11} : use their unified definitions.

Branch 3: $@ = 1$

Selecting A by an inequality condition

For given a and b , two points on the tangent satisfy $@ = 1$, one on each side of A_1 . To select a unique $A(X, Y)$ for a graph or calculation, specify one of the following conditions. Write $A_1 = (A_1X, A_1Y)$.

- 1) $X > A_1X, Y < A_1Y$
- 2) $X < A_1X, Y > A_1Y$
- 3) $X > A_1X, Y > A_1Y$
- 4) $X < A_1X, Y < A_1Y$

The two points lie on opposite sides of A_1 along the tangent. Thus a positive slope permits only 3) or 4), and a negative slope only 1) or 2).

Coincident Points and Special Cases

For a horizontal tangent, both points have the same y-coordinate as A_1 ; select using the x-condition alone. Horizontal and vertical reference lines are treated separately in Z2. Apply the same selection rule to points on a curve (function) where $@ = 1$.

$$A_7 = A_8 = A_9 = A_{9a} = A_{10} = A_{10}'$$

A_{11} follows the unified definition. Since $A_{10} = A_7$,
 $|AA_{10}| = |AA_7| = |AA_{11}|$.

Final point A_{12}

Vector $AA_{12} = \text{vector } AA_6 + \text{vector } AA_{11}$.

Special cases

1) $y = ax$, $A(x, ax)$, $b = 0$. $A_1 = O$, so A_7 is undefined. Use the direction AO , which bisects the two limiting directions as b approaches zero from either side, to define $A_{11} = O$. Then $A_{12} = A$ (complete self-return). This is an extension by convention, not continuity at $b = 0$.

2) $A = A_1$ (OA perpendicular to the tangent; $|AA_1| = 0$). $@$ is undefined and circle (1) has zero radius, so A_7 and subsequent points cannot be generated. A_{11} cannot be formed; use vector $AA_{12} = \text{vector } AA_6$, hence $A_{12} = A_6$.

3) The directions from A toward O and toward A_{10} are parallel. If they are identical or opposite, the direction of rotation for A_{11} is undetermined and requires a separate rule.

Z1: Graphing Rules

Basic setup

Mark the origin $O = (0, 0)$.

Use an axis scale ratio of 1:1 and a grid interval of 1.

Display the x- and y-axes in standard form.

Reference elements

Draw the tangent $y = ax + b$.

Mark $A(x, ax + b)$.

Mark A_1 , the perpendicular foot from O to the tangent.

Circles

Circle (1): center A_1 , radius $|A_1A|$.

Circle (2), $@ < 1$: center A_8 , radius $|A_8A_9|$.

Circle (2), $@ > 1$: center A_9 , radius $|A_9A_7|$.

Circle (2), $@ = 1$: undefined.

Point-label rules

Label O , A , A_1 , A_2 , A_{4a} , A_{4b} , A_{5a} , A_{5b} , A_6 , A_7 , A_8 , A_9 , A_{9a} , A_{10} , $A_{10'}$, A_{11} , A_{12} .

Do not label A_{3a} , $A_{3a'}$, A_{3b} , $A_{3b'}$, A_{3c} , A_{3d} .

Z1: Labels and Connecting Lines

Coordinate labels

Show coordinates only for A , $A1$, $A10'$, $A12$, to five decimal places. In a side panel list a , b , x , and $@$. For $@ = 1$, replace x with the selected inequality-condition number.

Basic connecting lines (always shown)

OA , $OA1$, $OA2$, $A1A2$, $A1A6$, $A1A7$, $AA2$, $AA6$, $AA9$, $AA10$, $AA11$, $AA12$, $A1A5a$, $A1A5b$, $A2A4a$, $A2A4b$, $A2A5a$, $A2A5b$, $A5aA6$, $A5bA6$, $A6A12$, $A11A12$, $A7A9$, $A7A9a$, $A8A9$, $A8A9a$, $A10A10'$, $A10'A11$.

Additional lines by branch

$@ < 1$: $AA8$, $A9A10$, $A9A10'$, $OA8$.

$@ > 1$: $AA7$, $A7A10$, $A7A10'$, $OA7$.

$@ = 1$: since $A7 = A8 = A9 = A9a = A10 = A10'$, label only $A10'$ and draw $AA10'$.

Special-case annotation

For $A = A1$, annotate: “ $@$ undefined ($A = A1$, $|AA1| = 0$)” and “ $A11$ cannot be generated $\Rightarrow AA12 = AA6 \Rightarrow A12 = A6$.” Omit circles (1) and (2), and $A7$ and subsequent points and connecting lines.

Z2 calculation method

Z2: Horizontal and Vertical Lines

Z2 applies when the reference line is vertical, $x = k$, or horizontal, $y = c$.

The point A has one of four forms:

$x = k$: $A = (k, k_1)$, or $A = (k, k)$.

$y = c$: $A = (c_1, c)$, or $A = (c, c)$.

Origin: $O = (0, 0)$.

A1 (same rule throughout)

The perpendicular foot from O to the reference line:

$x = k \Rightarrow A1 = (k, 0)$.

$y = c \Rightarrow A1 = (0, c)$.

A2 (axis projection of A)

$x = k, A = (k, k_1) \Rightarrow A2 = (0, k_1)$.

$x = k, A = (k, k) \Rightarrow A2 = (0, k)$.

$y = c, A = (c_1, c) \Rightarrow A2 = (c_1, 0)$.

$y = c, A = (c, c) \Rightarrow A2 = (c, 0)$.

Circle (1) and A6

Circle (1): center A1, radius $|A1A|$.

A6: intersection of the reference line with the line through A2 perpendicular to A1A2.

Z2: Branch Selection and A7-A12

The reference line is vertical for $x = k$ and horizontal for $y = c$. The definition of $@$ is the same as in Z1. The subsequent stages for $@ < 1$, $@ > 1$, and $@ = 1$ use the same branch rules as Z1.

Selecting a condition when $@ = 1$

For $x = k$, $A = (k, k_1)$ and $A_1 = (k, 0)$ share the same x-coordinate. $@ = 1$ means $|k_1| = |k|$. Choose either $Y > A_1Y$ ($k_1 > 0$) or $Y < A_1Y$ ($k_1 < 0$).

For $y = c$, $A = (c_1, c)$ and $A_1 = (0, c)$ share the same y-coordinate. $@ = 1$ means $|c_1| = |c|$. Choose either $X > A_1X$ ($c_1 > 0$) or $X < A_1X$ ($c_1 < 0$).

Thus the four conditions in Z1 reduce to two in Z2.

A7-A12: core rule

From A7 onward, perform the calculations exactly as in Z1: A7, A8, A9, A9a, A10, A10', A11, and the final vector sum for A12.

Only the reference line differs. Rotation, circle (2), and vector-sum rules remain those of Z1. Apply the same unified definition of A11: a 90° rotation using the direction from A toward O as reference.

Z2: Consequences and Graphing

Special consequence

A line $y = ax$ has a defined slope and belongs to Z1. Its self-return result ($A12 = A$ when $b = 0$) is given in Z1 special case 1 and is not treated separately in Z2.

Basic graph setup

Show the x- and y-axes and $O(0, 0)$. Use a 1:1 axis scale ratio and grid interval 1.

Reference line and points

Reference line: $x = k$ or $y = c$.

Points: A, A1, A2, A6, A7, A8, A9, A9a, A10, A10', A11, A12.

Circles

Circle (1): center A1, radius $|A1A|$.

Circle (2), $@ < 1$: center A8, radius $|A8A9|$.

Circle (2), $@ > 1$: center A9, radius $|A9A7|$.

Circle (2), $@ = 1$: omit.

Basic connecting lines

OA, A1A2, AA2, A2A6, AA6, A6A12, AA9, AA10, AA11, AA12, A7A9, A7A9a, A8A9, A8A9a, A11A12, A10A10', A10'A11.

Z2: Final Display Rules

Additional lines by branch

@ < 1: AA8, A9A10, A9A10', OA8.

@ > 1: AA7, A7A10, A7A10', OA7.

@ = 1: since $A7 = A8 = A9 = A9a = A10 = A10'$, label only A10' and draw AA10'.

Coordinates and annotations

Show coordinates only for A, A1, A10', A12.

Use five decimal places.

In a side panel show k or c, @, the coordinates of A12, |AA12|, and the angle of vector AA12. For @ = 1, also show the selected condition number.

Detailed Sangbun-vector calculations follow.

Detailed Calculations: A1 to A12

[Definitive order: definition → formula → reasoning]

Calculate A1 through A12 in construction order. Each stage gives the definition, formula, derivation, and a check.

Keep the detailed point names A3a, A3b, A4a, A4b, etc. No separate points named A3, A4, or A5 are introduced. After A10, calculate A10', followed by A11 and A12.

Basic inputs and frequently used quantities:

$$O = (0, 0), \quad A = (x, y_A), \quad y_A = ax + b$$

$$K = 1 + a^2, \quad P = Kx + ab$$

Two direction vectors shorten the calculations; they are not unit vectors.

$$T = (1, a), \quad N = (-a, 1)$$

$$T \cdot N = 0, \quad |T|^2 = |N|^2 = K$$

The general formulas apply to nondegenerate constructions on lines that are neither horizontal nor vertical. Axis-aligned lines and special cases are treated later. Proofs and figures for the two earlier problems are not included.

Tools: Line Intersections and Rotation

[Definitions of common operations]

For vectors $v = (v_x, v_y)$ and $w = (w_x, w_y)$, define Δ as the planar determinant.

$$\Delta(v, w) = v_x w_y - v_y w_x$$

The intersection of $p + \ell v$ and $q + rw$ is calculated below. Here p and q denote coordinates of existing points, not new construction points.

$$\mathcal{I}(p, v; q, w) = p + \frac{\Delta(q - p, w)}{\Delta(v, w)} v$$

Reason: take the determinant with w on both sides of $p + \ell v = q + rw$; the r -term vanishes.

$$\ell \Delta(v, w) = \Delta(q - p, w)$$

Require $\Delta(v, w) \neq 0$. A zero denominator means the lines are parallel or coincident; this formula cannot specify one intersection.

J denotes a counterclockwise rotation through 90° .

$$J(v_x, v_y) = (-v_y, v_x)$$

$$v \cdot Jv = 0, \quad |Jv| = |v|$$

Every subsequent I and J denotes the explicit arithmetic operation above.

Steps 1-2: A1 and A2

[Calculating perpendicular feet]

A1 is the perpendicular foot from O to $y = ax + b$. The direction from O to A1 is parallel to N.

$$A_1 = \frac{b}{K}N = \left(-\frac{ab}{K}, \frac{b}{K}\right)$$

Check: A1 satisfies the line equation, and N is perpendicular to the tangent direction T.

A2 is the perpendicular foot from O to the normal through A. That normal is $A + rN$; its perpendicular from O has direction T.

$$A_2 = A - \frac{A \cdot N}{K}N$$

$$A \cdot N = b, \quad A \cdot T = P$$

$$A_2 = \frac{P}{K}T = \left(\frac{P}{K}, \frac{aP}{K}\right)$$

Distinguish the two feet: A1 lies on the tangent; A2 lies on the normal through A.

$$A = A_1 + A_2$$

The last identity adds coordinates with $O = (0, 0)$. It can be used to check later calculations.

Step 3: Auxiliary Points in the A3 Family

[Intersections with the axes, normal, and tangent]

The normal at A is $(x, y_A) + r(-a, 1)$. Impose the condition for intersection with each coordinate axis.

$$A_{3a} = (P, 0), \quad A_{3b} = \left(0, \frac{P}{a}\right)$$

A3a: $y = 0$ gives $r = -y_A$, so its x-coordinate is $x + ay_A = P$. A3b: $x = 0$ gives $r = x/a$, so its y-coordinate is $y_A + x/a = P/a$.

Form A3a' and A3b' by the prescribed coordinate combinations.

$$A'_{3a} = \left(P, \frac{b}{K}\right)$$

$$A'_{3b} = \left(-\frac{ab}{K}, \frac{P}{a}\right)$$

A3c is where the vertical through A3a meets the tangent. A3d is where the horizontal through A3b meets the tangent.

$$A_{3c} = (P, aP + b)$$

$$A_{3d} = \left(\frac{P - ab}{a^2}, \frac{P}{a}\right)$$

This requires $a \neq 0$. In Z_2 , horizontal and vertical lines bypass the A3 family.

Step 4: A4a and A4b

[Calculating intersections of two lines]

A4a is the intersection of OA3c and AA3a'. A4b is the intersection of OA3d and AA3b'.

$$A_{4a} = \mathcal{I}(O, A_{3c}; A, A'_{3a} - A)$$

$$A_{4b} = \mathcal{I}(O, A_{3d}; A, A'_{3b} - A)$$

Since $O = (0, 0)$, the arithmetic expressions simplify further.

$$A_{4a} = \frac{\Delta(A, A'_{3a} - A)}{\Delta(A_{3c}, A'_{3a} - A)} A_{3c}$$

$$A_{4b} = \frac{\Delta(A, A'_{3b} - A)}{\Delta(A_{3d}, A'_{3b} - A)} A_{3d}$$

Procedure: first find the A3 coordinates, then calculate the direction-vector determinant and numerator determinant, and divide.

Check that A4a lies on both OA3c and AA3a'. Check A4b in the same way.

Regardless of the heading or name, this stage uses exactly the definitive text's definition as an intersection of two lines.

Step 5: A5a and A5b

[Intersections with the perpendicular through A2]

Construct line (1) through A2 perpendicular to A1A2. Let its direction vector be w .

$$w = J(A_2 - A_1)$$

Intersect line (1) with the extended lines A1A4a and A1A4b, respectively.

$$A_{5a} = \mathcal{I}(A_2, w; A_1, A_{4a} - A_1)$$

$$A_{5b} = \mathcal{I}(A_2, w; A_1, A_{4b} - A_1)$$

In general position, simplifying the intersection formulas gives the direct expressions below. The quantities r_a and r_b are auxiliary scalars.

$$r_a = \frac{b}{KP + ab}, \quad r_b = \frac{a^2b}{KP - ab}$$

$$A_{5a} = \frac{(P + br_a)T + Pr_aN}{K}$$

$$A_{5b} = \frac{(P + br_b)T + Pr_bN}{K}$$

Check that each point lies on line (1) through A2 and on A1A4a or A1A4b. Even if a direct formula is finite, the preceding constructions must be defined.

Step 6: A6

[A vector sum based at A1]

The definition is the vector sum of A1A5a and A1A5b.

$$A_6 - A_1 = (A_{5a} - A_1) + (A_{5b} - A_1)$$

$$A_6 = A_{5a} + A_{5b} - A_1$$

Introduce the auxiliary scalar h to shorten the formula for A6.

$$h = r_a + r_b = b \left(\frac{1}{KP + ab} + \frac{a^2}{KP - ab} \right)$$

$$A_6 = \frac{(2P + bh)T + (Ph - b)N}{K}$$

Adding the T-coefficients gives $2P + bh$. The N-coefficients sum to Ph ; subtracting $A_1 = bN/K$ once gives $Ph - b$.

For direct coordinate calculation, use:

$$x_6 = \frac{2P + bh - a(Ph - b)}{K}$$

$$y_6 = \frac{a(2P + bh) + Ph - b}{K}$$

The coordinate difference between this expression and the intersection-based A6 simplifies exactly to (0, 0): a rational-function identity in general position.

Step 7: Circle (1) and A7

[The direction from O through and beyond A1]

Circle (1) has center A1 and radius |A1A|.

$$R = |A - A_1| = \frac{|P|}{\sqrt{K}}$$

A7 lies beyond A1 on the ray from O through A1, at its intersection with circle (1). Move one radius from A1 away from O.

$$A_7 = A_1 - R \frac{O - A_1}{|O - A_1|}$$

For $b \neq 0$, the coordinate expression simplifies to:

$$A_7 = A_1 + \frac{\text{sign}(b)|P|}{K} N$$

Two checks:

$$|A_7 - A_1| = R$$

O, A1, and A7 are collinear; viewed from A1, O and A7 lie in opposite directions.

If $b = 0$, $A_1 = O$ and the direction is undefined. If $P = 0$, circle (1) has zero radius. Use the special rules in either case.

Step 8: A8

[The second intersection with circle (1)]

By definition, AA8 is perpendicular to OA. Since $O = (0, 0)$, vector OA equals A. Thus choose the line direction $v = J(A)$.

$$v = J(A), \quad A_8 = A + \ell v$$

Impose the condition that A8 lies on circle (1):

$$|A + \ell v - A_1|^2 = R^2$$

Since $|A - A_1|^2 = R^2$, expansion gives two roots:

$$\ell(2(A - A_1) \cdot v + \ell|v|^2) = 0$$

The root $\ell = 0$ is the known point A. Choose the other root.

$$\ell = -\frac{2(A - A_1) \cdot v}{|v|^2}$$

$$A_8 = A - \frac{2(A - A_1) \cdot J(A)}{|A|^2} J(A)$$

Check $|A_8 - A_1| = R$ and $(A_8 - A) \cdot A = 0$. General position, with a second intersection distinct from A, is assumed.

Step 9: A9 and A9a

[Choosing the inside and outside candidates]

Find the midpoint M and direction vector d of A_7A_8 . M is auxiliary notation for the midpoint, not a new numbered A-point.

$$M = \frac{A_7 + A_8}{2}, \quad d = A_8 - A_7$$

Move from the midpoint in either perpendicular direction by half the segment length to obtain two candidates:

$$C_+ = M + \frac{1}{2}J(d)$$

$$C_- = M - \frac{1}{2}J(d)$$

Compare each candidate's distance to the center A_1 of circle (1).

$$|C - A_1| < R \implies A_9 = C$$

$$|C - A_1| > R \implies A_{9a} = C$$

Compare squared distances with R^2 to avoid square roots. Follow the rule: A_9 is inside and A_{9a} is outside.

For $t = |b|/|P| = 1$, $A_7 = A_8$ and the candidates coincide. Apply the coincident-point rule: $A_9 = A_{9a} = A_7$.

Step 10: Circle (2) and A10

[Choose the reference point and center by branch]

$$t = \frac{|b|}{|P|}$$

F and C are auxiliary names for existing points. F is the reference point (the perpendicular foot), and C is the center of circle (2).

Branch	F: reference point	C: center
$0 < t < 1$	A9	A8
$t > 1$	A7	A9

Circle (2) has radius $|F - C|$. Let w be the direction of the line through F perpendicular to AF.

$$w = J(F - A), \quad A_{10} = F + qw$$

Substitution into circle (2) gives F at $q = 0$ and this second intersection:

$$q = -\frac{2(F - C) \cdot w}{|w|^2}$$

$$A_{10} = F - \frac{2(F - C) \cdot w}{|w|^2} w$$

Expand $|F + qw - C|^2 = |F - C|^2$, factor out q , and choose the nonzero root. This applies when the second intersection is distinct.

After Step 10: A10'

[Calculating the endpoint of the hypotenuse]

A10' is the intersection of the line through A10 perpendicular to AA10 with the extended line from A toward F.

$$d = F - A, \quad z = A_{10} - A$$

Write $A_{10}' = A + rd$. The perpendicularity of $A_{10}A_{10}'$ and AA_{10} gives:

$$(A + rd - A_{10}) \cdot z = 0$$

$$r(d \cdot z) = |z|^2$$

Since $A_{10} - F$ is perpendicular to d , $d \cdot z = |d|^2$. Therefore:

$$A'_{10} = A + \frac{|A_{10} - A|^2}{|F - A|^2} (F - A)$$

Check that A , F , A_{10}' are collinear and AA_{10} is perpendicular to $A_{10}A_{10}'$. F is the perpendicular foot from A_{10} to the hypotenuse AA_{10}' .

At $t = 1$, apply the separate coincident-point rule directly.

$$A_7 = A_8 = A_9 = A_{9a} = A_{10} = A'_{10}$$

Step 11: A11

[Rotate AA10 through 90° about A]

First calculate $z = A_{10} - A$. Because the center of rotation is A, add the rotated vector back to A.

$$z = A_{10} - A, \quad s = \text{sign} \left(-\frac{b}{p} \right)$$

The coordinate formulas with the prescribed rotation directions are:

$$A_{11} = A + sJ(z) \quad (0 < t \leq 1)$$

$$A_{11} = A - sJ(z) \quad (t > 1)$$

In general position, s is the sign of rotation from AO toward AA10. Use that direction for $t \leq 1$ and the opposite direction for $t > 1$.

$$|A_{11} - A| = |A_{10} - A|$$

$$(A_{11} - A) \cdot (A_{10} - A) = 0$$

To calculate directly without A10, use the rotational components X and Y.

$$x_{11} = x + \frac{P}{K}(X - asY)$$

$$y_{11} = ax + b + \frac{P}{K}(aX + sY)$$

X and Y for the Direct A11 Formula

[The two prescribed pieces, without reversal]

For $t > 1$:

$$X = -\frac{2}{1+t^2}, \quad Y = \frac{2t^2}{1+t^2}$$

For $0 < t \leq 1$:

$$D = (1+t^2)(5t^2-2t+1)$$

$$X = -\frac{t(3t^3-15t^2+9t-5)}{D}$$

$$Y = -\frac{14t^4-9t^3+5t^2-3t+1}{D}$$

At $t = 1$, use the lower piece:

$$(X, Y) = (1, -1)$$

X and Y are components of A11 - A in normalized coordinates with A1 as origin and A at (1, 0). Restore actual coordinates as follows:

$$A_{11} - A = \frac{P}{K}[XT + sYN]$$

This gives the same result as the sequential rotation formulas on the preceding page. Compare the two methods as a check.

Step 12: A12

[Sum of two vectors starting at A]

The final stage is the vector sum of AA6 and AA11.

$$A_{12} - A = (A_6 - A) + (A_{11} - A)$$

$$A_{12} = A_6 + A_{11} - A$$

Once A6 and A11 are known, add their coordinates and subtract the coordinates of A once.

$$X_{12} = X_6 + X_{11} - X$$

$$Y_{12} = Y_6 + Y_{11} - (ax + b)$$

A compact direct formula for the full calculation is:

$$U = P(2 + X) + bh$$

$$V = P(h + sY) - b$$

$$A_{12} = \left(\frac{U - aV}{K}, \frac{aU + V}{K} \right)$$

Here h is the scalar from the A6 stage. Adding the T- and N-coefficients in A6 + (A11 - A) gives U and V, so the sequential and compact formulas agree.

Horizontal and Vertical Lines: Z2

[First apply the Z2 definition of A6]

For horizontal and vertical lines, Z2 defines A1, A2, and A6 directly. Do not force the Z1 construction of A3-A5 onto these cases.

Horizontal line $y = c$, $A = (x, c)$, $cx \neq 0$:

$$A_1 = (0, c), \quad A_2 = (x, 0)$$

$$A_6 = \left(x + \frac{c^2}{x}, c\right)$$

$$t = |c/x|, \quad s = \text{sign}(-c/x)$$

$$A_{11} = (x + xX, c + xsY)$$

$$A_{12} = \left(x + \frac{c^2}{x} + xX, c + xsY\right)$$

Vertical line $x = k$, $A = (k, y)$, $ky \neq 0$:

$$A_1 = (k, 0), \quad A_2 = (0, y)$$

$$A_6 = \left(k, y + \frac{k^2}{y}\right)$$

$$t = |k/y|, \quad s = \text{sign}(k/y)$$

$$A_{11} = (k - ysY, y + yX)$$

$$A_{12} = \left(k - ysY, y + \frac{k^2}{y} + yX\right)$$

Special Cases and Preliminary Checks

[Special rules take precedence over general formulas]

1. A general line with $b = 0$: $A1 = O$, so the direction of $A7$ is undefined. If $A \neq O$, use the special convention $A11 = O$ and $A12 = A$.
2. $P = 0$, or $A = A1$: circle (1) has zero radius and t is undefined. Do not construct $A11$; if $A6$ is defined, use $A12 = A6$.
3. $t = 1$: $A7 = A8 = A9 = A9a = A10 = A10'$. Use this coincidence rule instead of finding a second intersection with circle (2), then rotate to obtain $A11$.
4. A zero intersection determinant: parallel or coincident lines may not have a unique intersection. Do not assign an arbitrary value without a separate rule.
5. $KP + ab = 0$ or $KP - ab = 0$: the compact $A5$, $A6$, and $A12$ formulas have zero denominators. Exclude these cases from the general formulas.
6. If the rotation direction is undetermined, or $A = O$, consult the special rules. Do not automatically extend the general formulas.

Source basis: the attached Sangbunhak definitive revised text v20 ($A10'$ unified definition, 260914 / 261002_003609) and the user-provided $A11$ component formulas. Decimals in the following tables are rounded from exact rational calculations.

Example: A1 through A5

[Sequential calculation check: five decimal places]

$$y = \frac{3}{5}x + 6, \quad A = \left(\frac{180}{17}, \frac{210}{17}\right)$$

$$K = \frac{34}{25}, \quad P = 18, \quad t = \frac{1}{3}, \quad s = -1$$

Point	x-coordinate	y-coordinate
A1	-2.64706	4.41176
A2	13.23529	7.94118
A3a	18.00000	0.00000
A3a'	18.00000	4.41176
A3b	0.00000	30.00000
A3b'	-2.64706	30.00000
A3c	18.00000	16.80000
A3d	40.00000	30.00000
A4a	11.82060	11.03256
A4b	12.70588	9.52941
A5a	12.48115	11.33484
A5b	12.87018	9.58418

The A3 axis intersections, A4 line intersections, and A5 perpendicular-line intersections were computed in order. The direct A5 formulas agree.

Example: A6 through A12

[Exact results and decimal representations]

Point	x-coordinate	y-coordinate
A6	27.99839	16.50726
A7	-10.58824	17.64706
A8	3.17647	18.70588
A9	-3.17647	11.29412
A9a	-4.23529	25.05882
A10	-4.23529	25.05882
A10'	-16.94118	10.23529
A11	23.29412	27.17647
A12	40.70427	31.33078

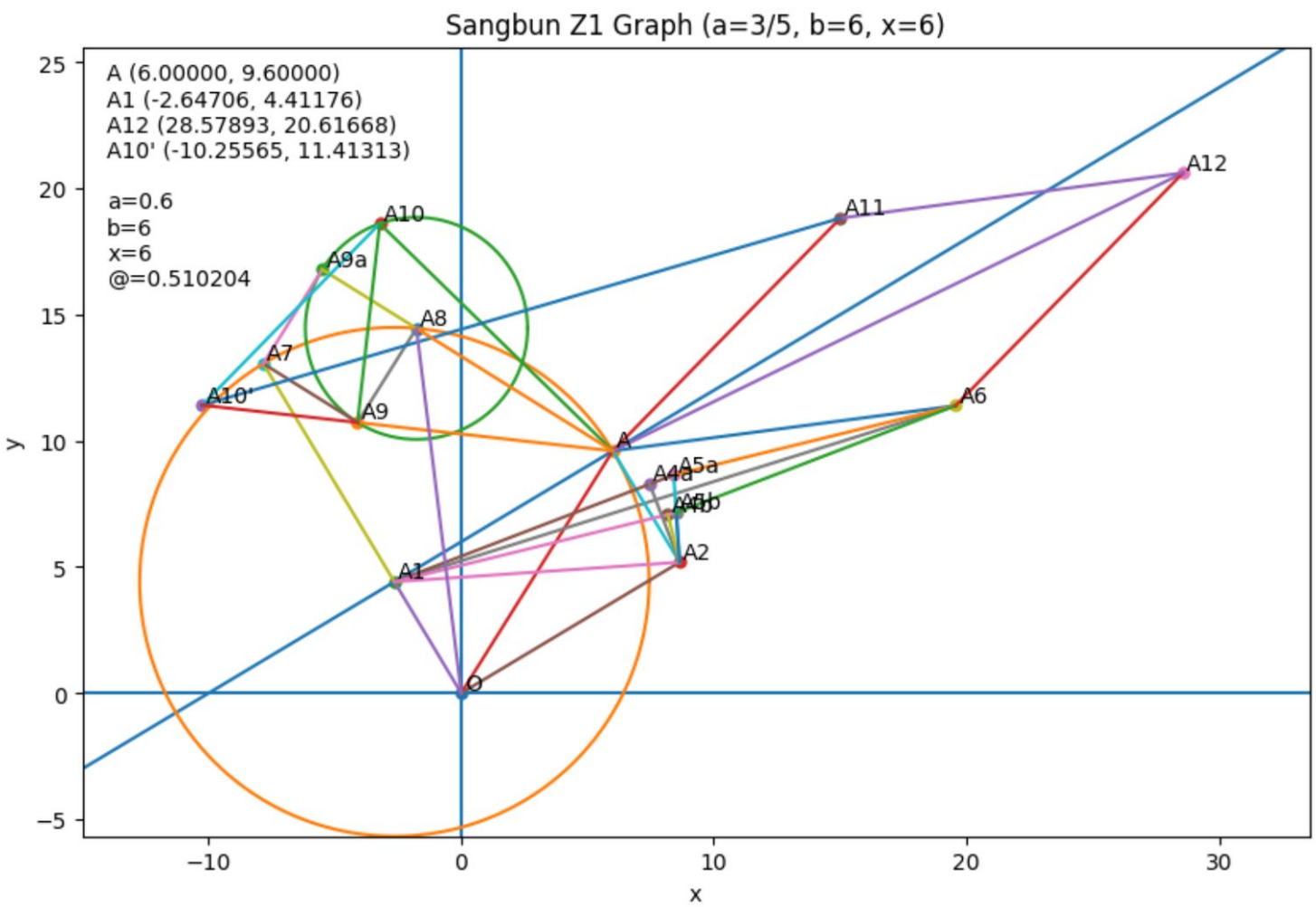
Exact rational coordinates of A11 and A12:

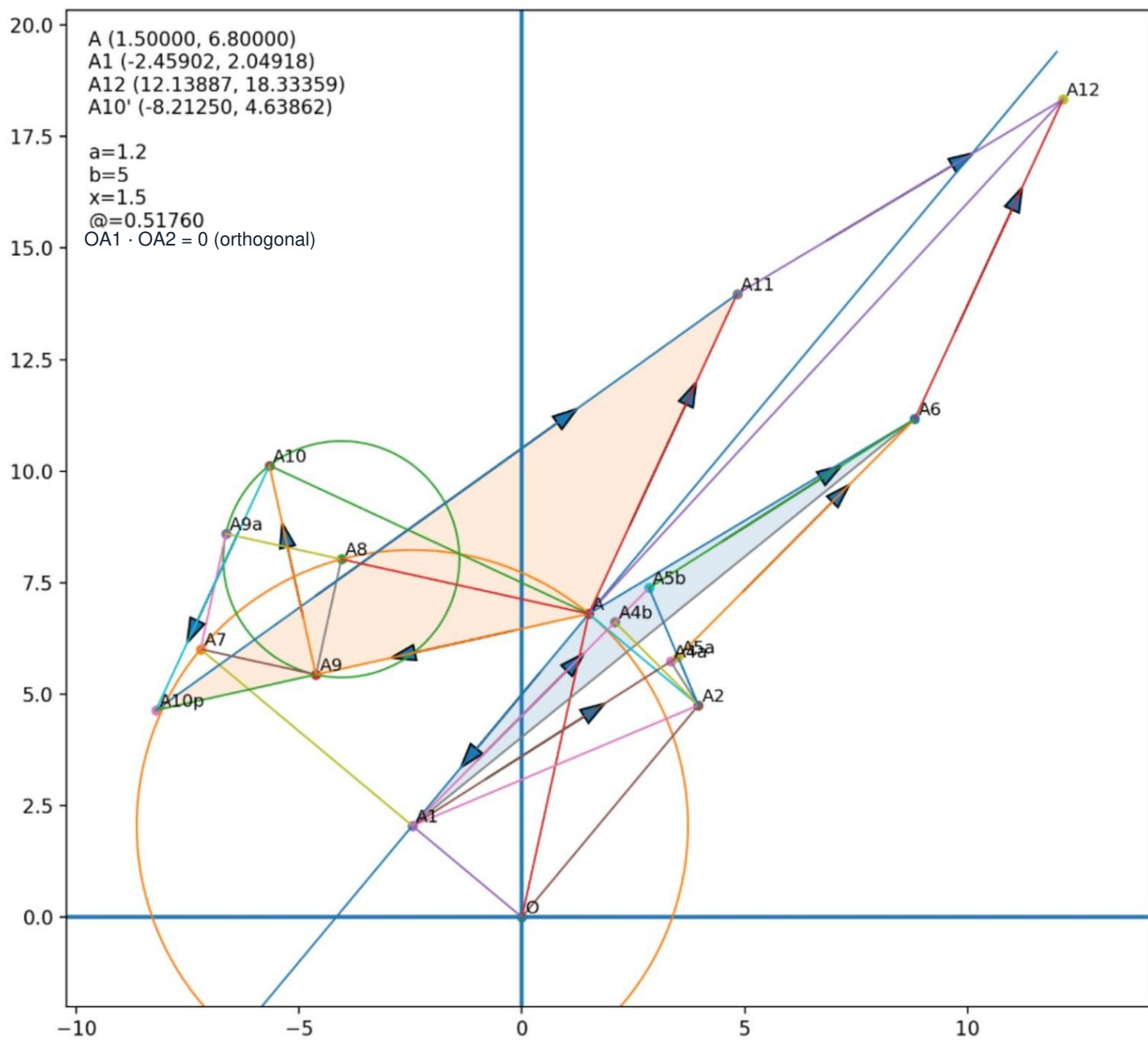
$$A_{11} = \left(\frac{396}{17}, \frac{462}{17} \right)$$

$$A_{12} = \left(\frac{782621}{19227}, \frac{200799}{6409} \right)$$

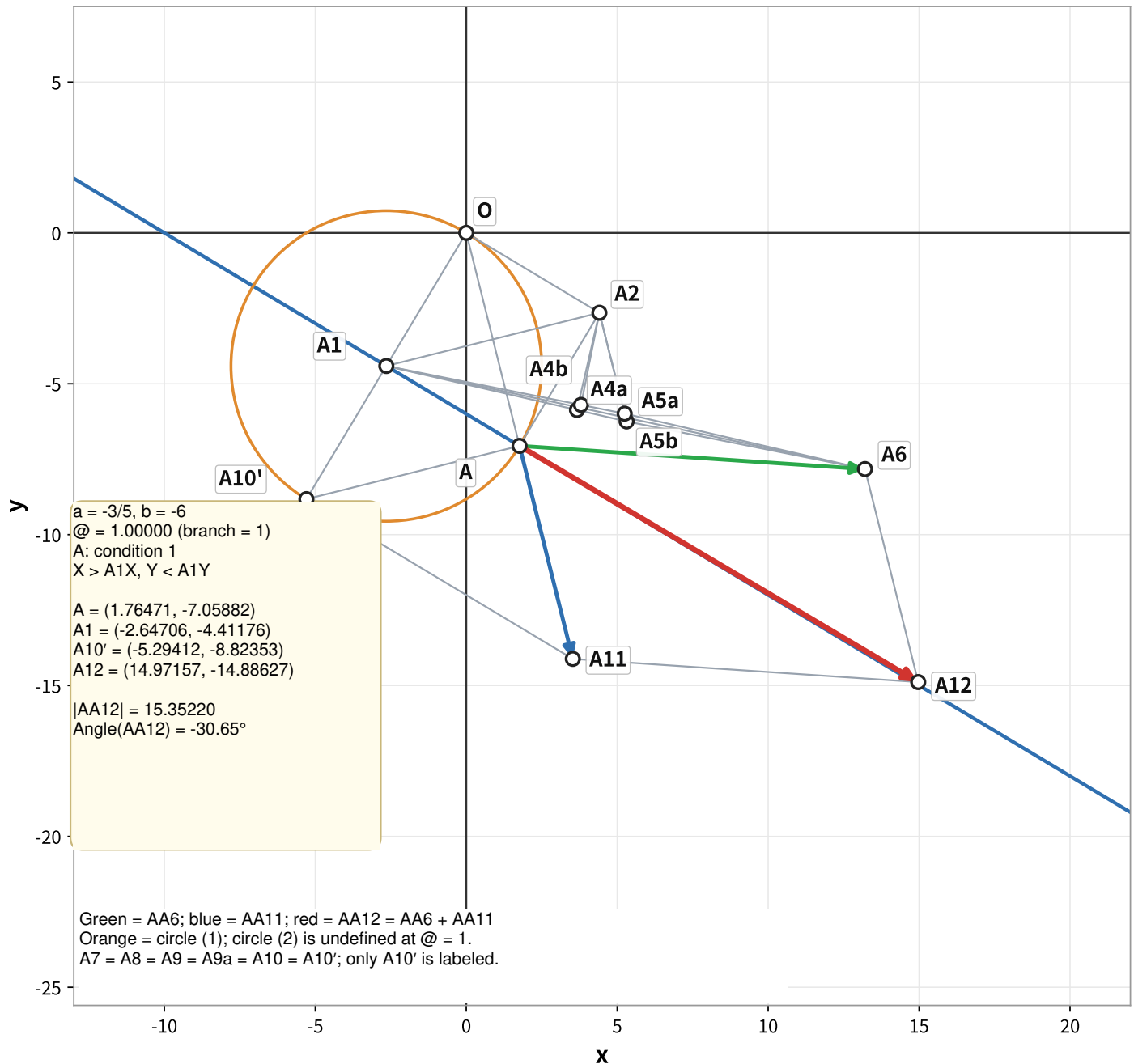
Sequential intersections starting from A1 were checked against the direct A6, A11, and A12 formulas. Each coordinate difference is exactly zero.

Sangbun Vector Graphs

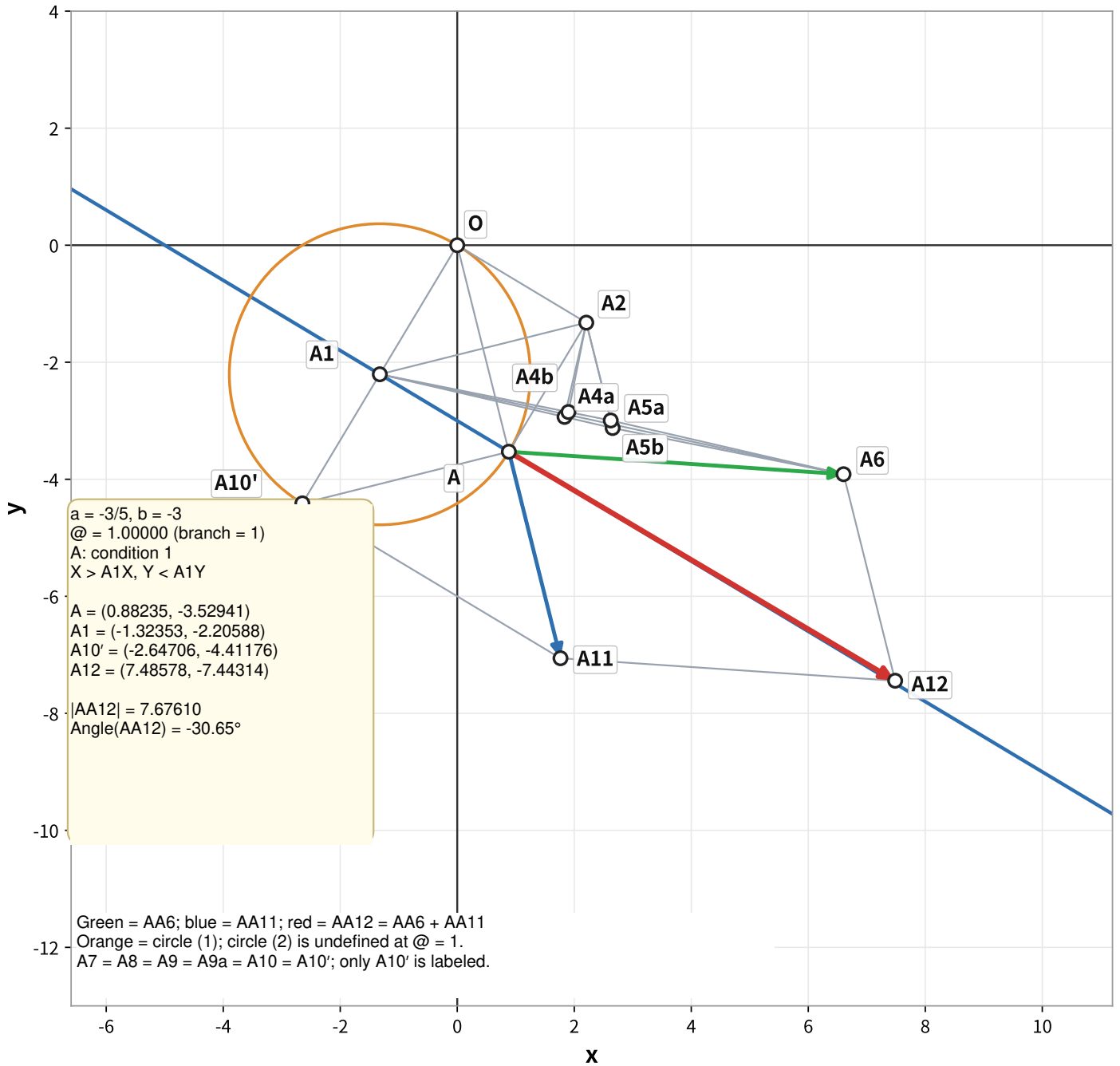




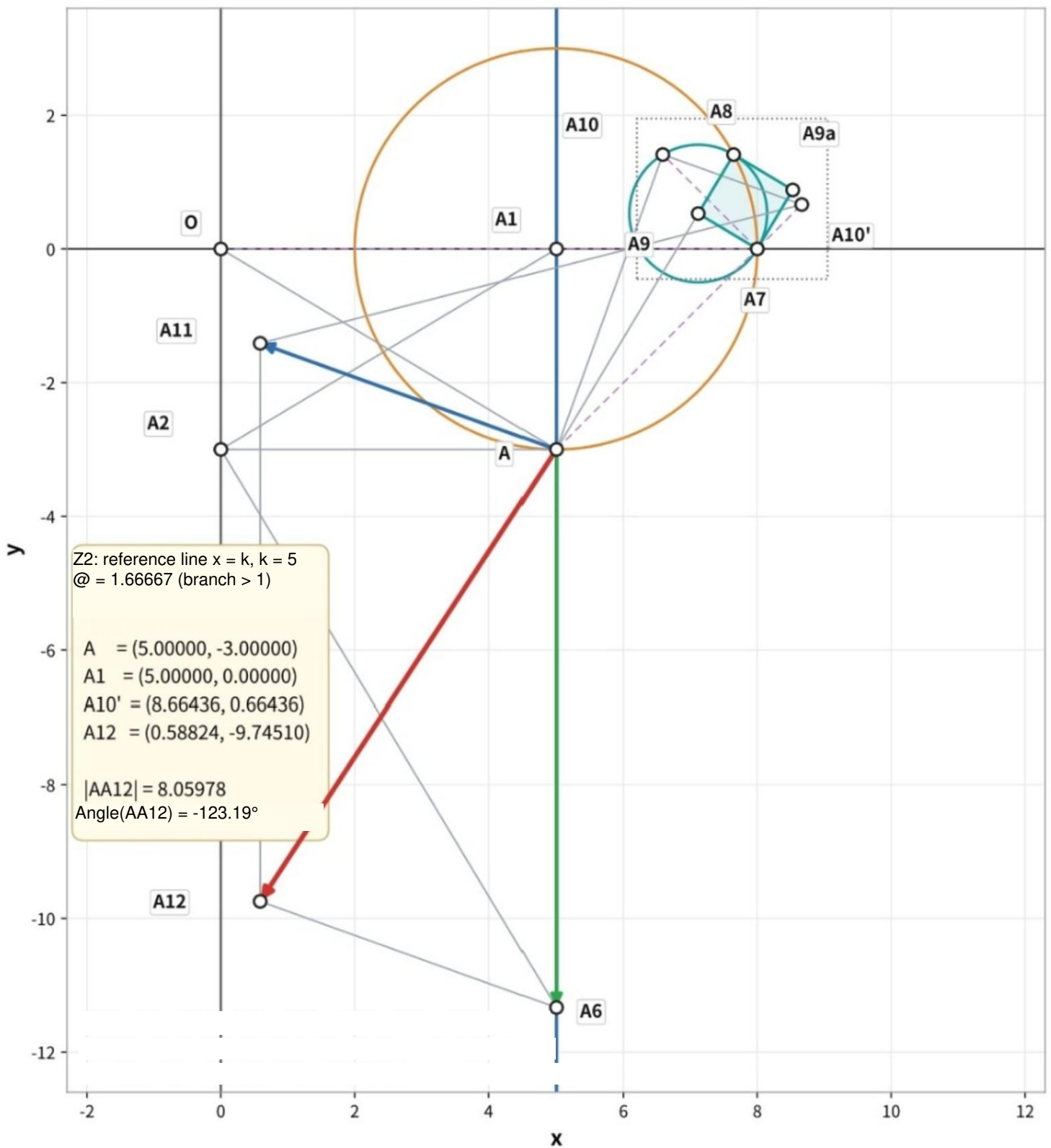
Z1 Sangbun Vector: $a = -3/5$, $b = -6$, $@ = 1$ (condition 1)



Z1 Sangbun Vector: $a = -3/5$, $b = -3$, $@ = 1$ (condition 1)

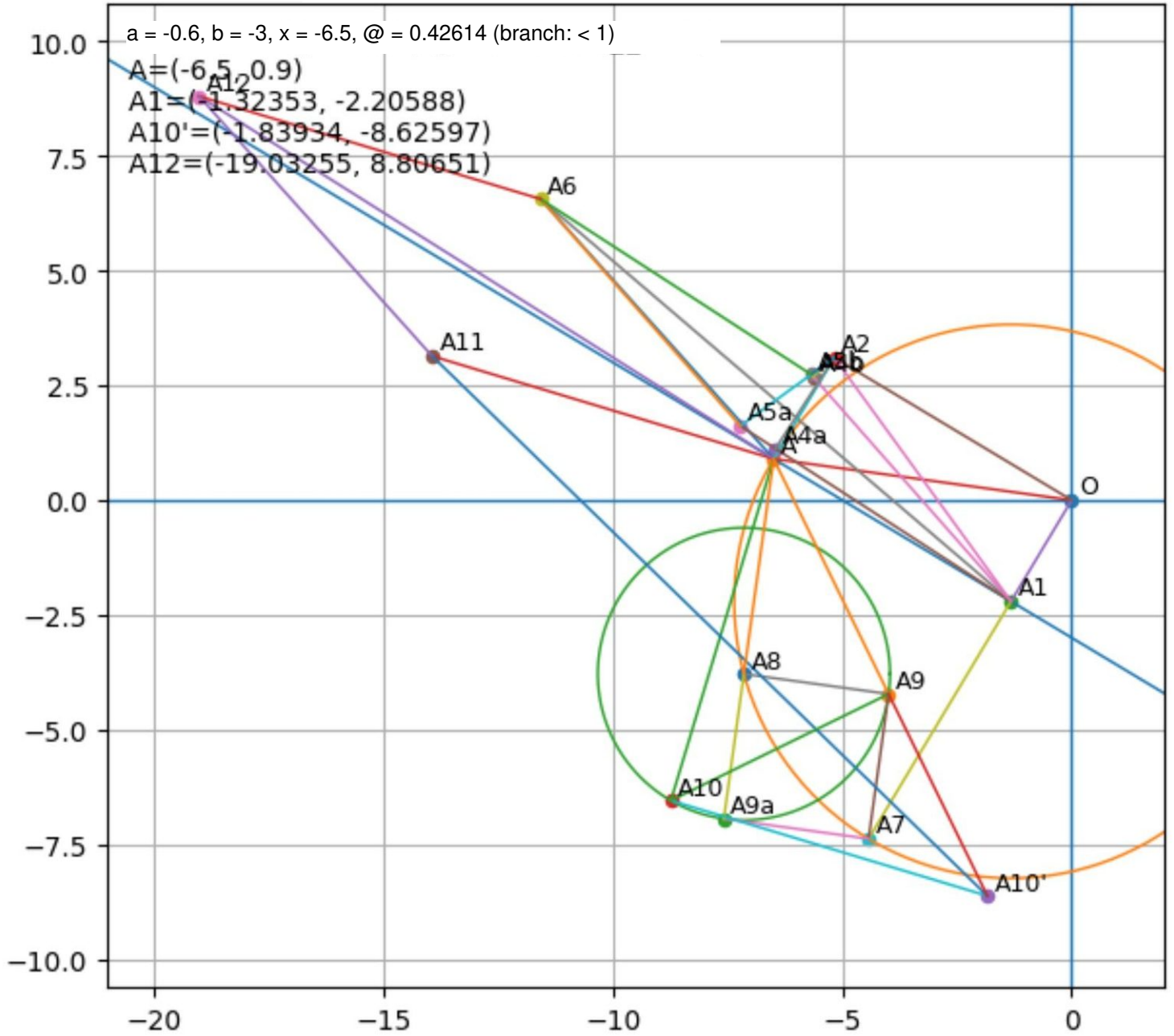


Z2 Sangbun Vector: $x = 5$, $A = (5, -3)$, $@ > 1$

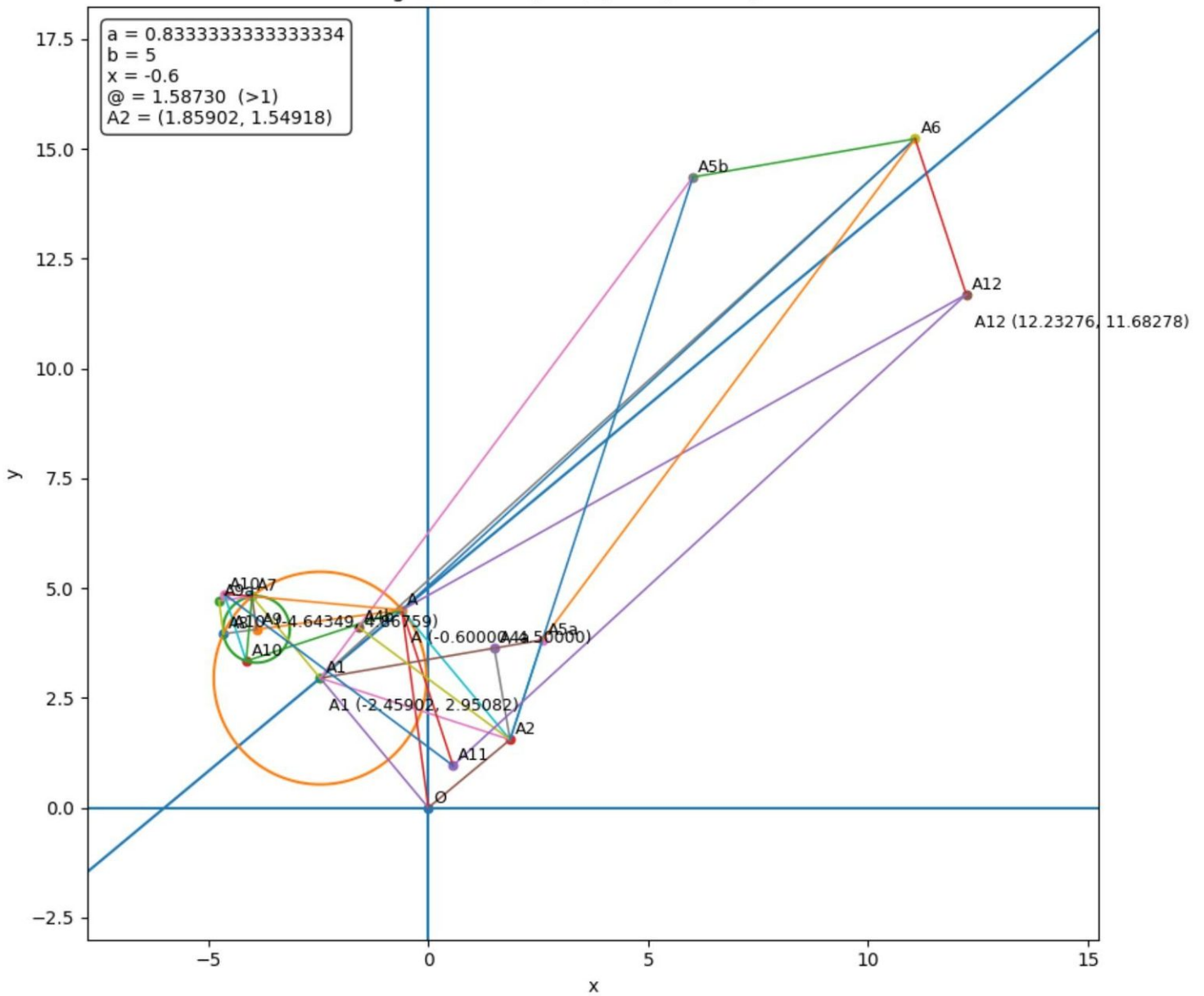


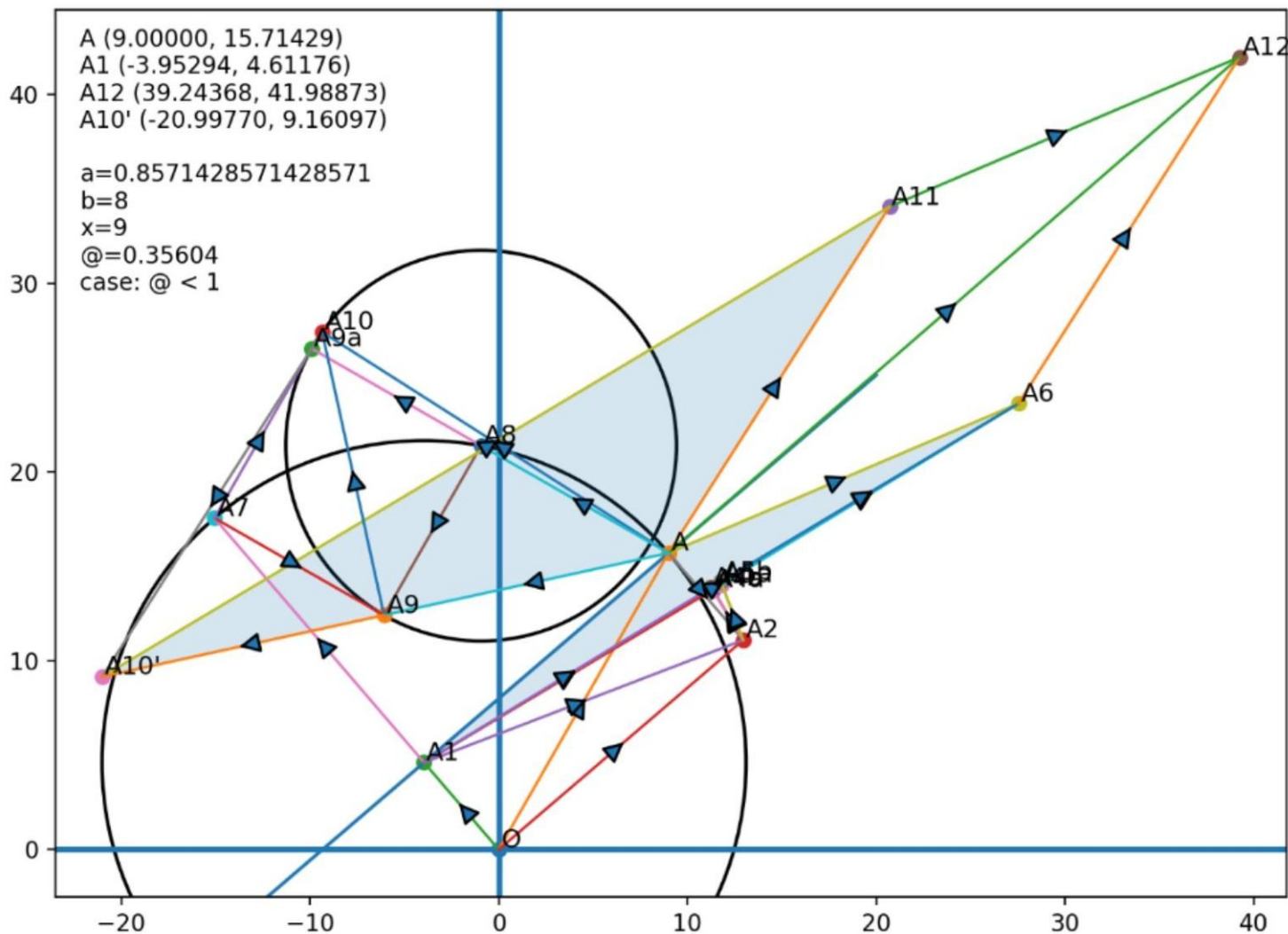
Green = AA6; blue = AA11; red = AA12 = AA6 + AA11
 Orange = circle (1); green = circle (2). Teal square: A7 A9 A8 A9a.
 Purple dashed lines: additional connections for $@ > 1$.

Z1 Sangbun Vector Graph ($a = -3/5$, $b = -3$, $x = -6.5$)

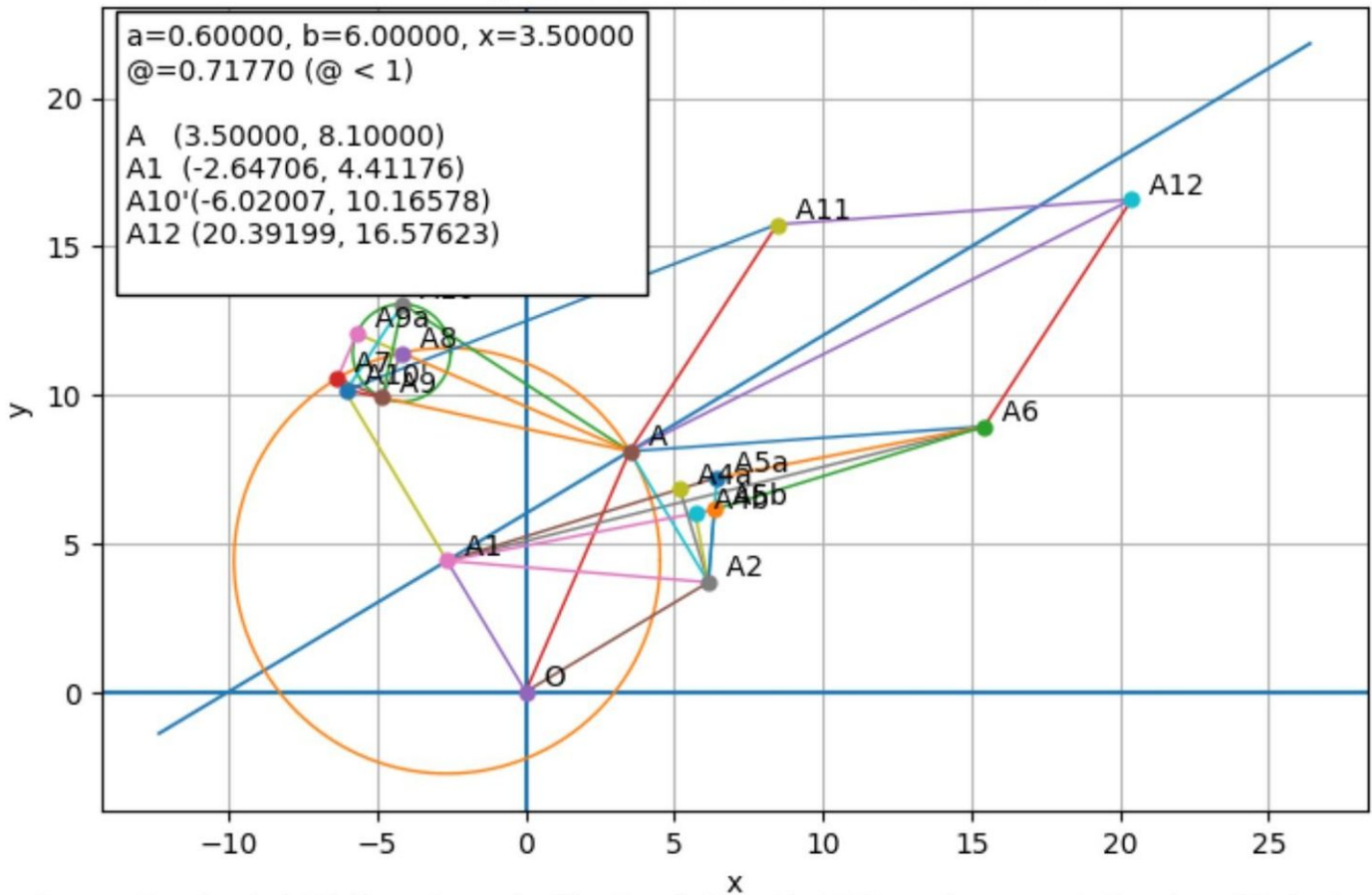


Z1 Sangbun Vector ($a=5/6$, $b=5$, $x=-0.6$) - A2 fixed



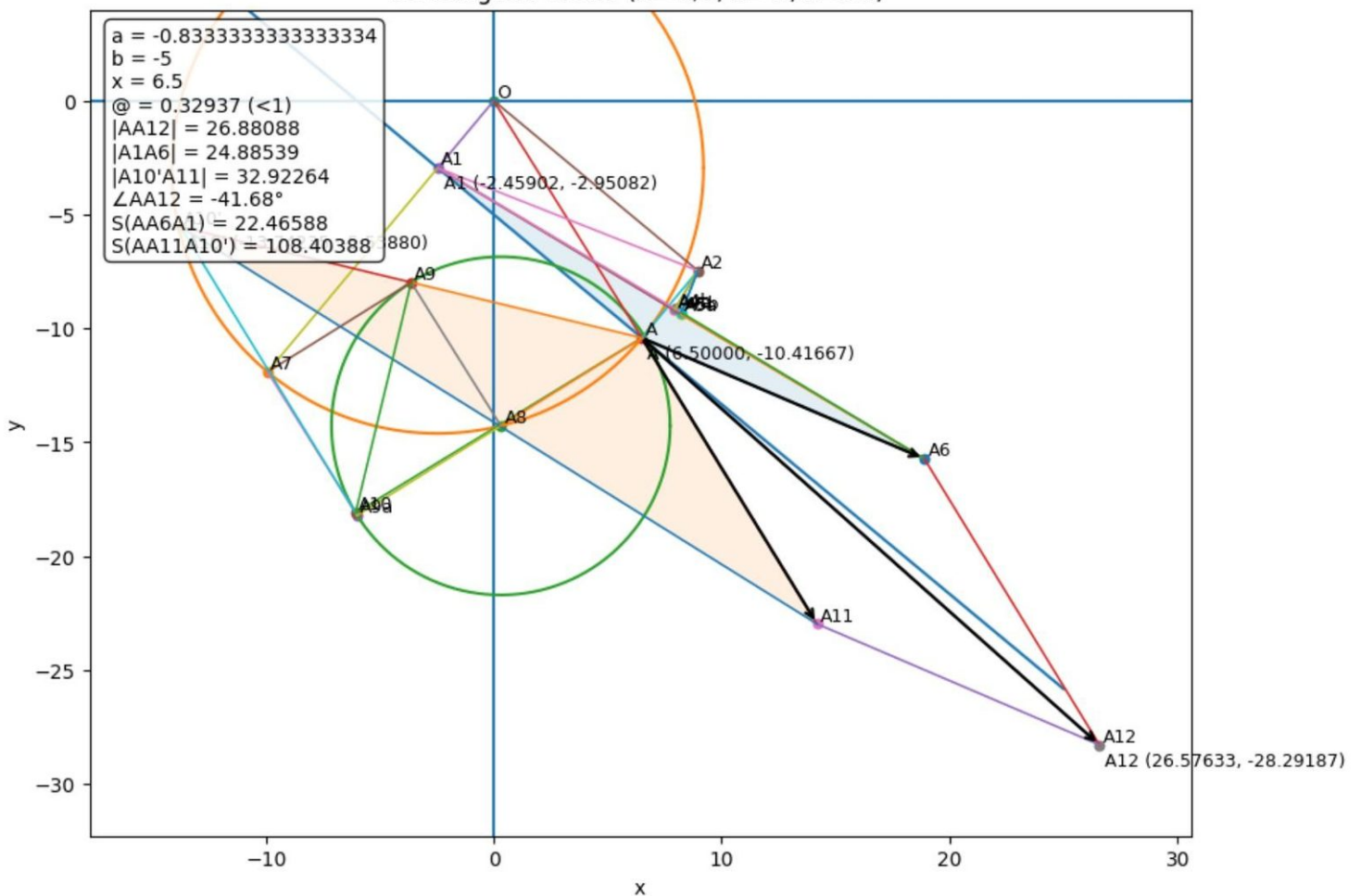


Sangbunhak Z1 ($a=3/5$, $b=6$, $x=3.5$)



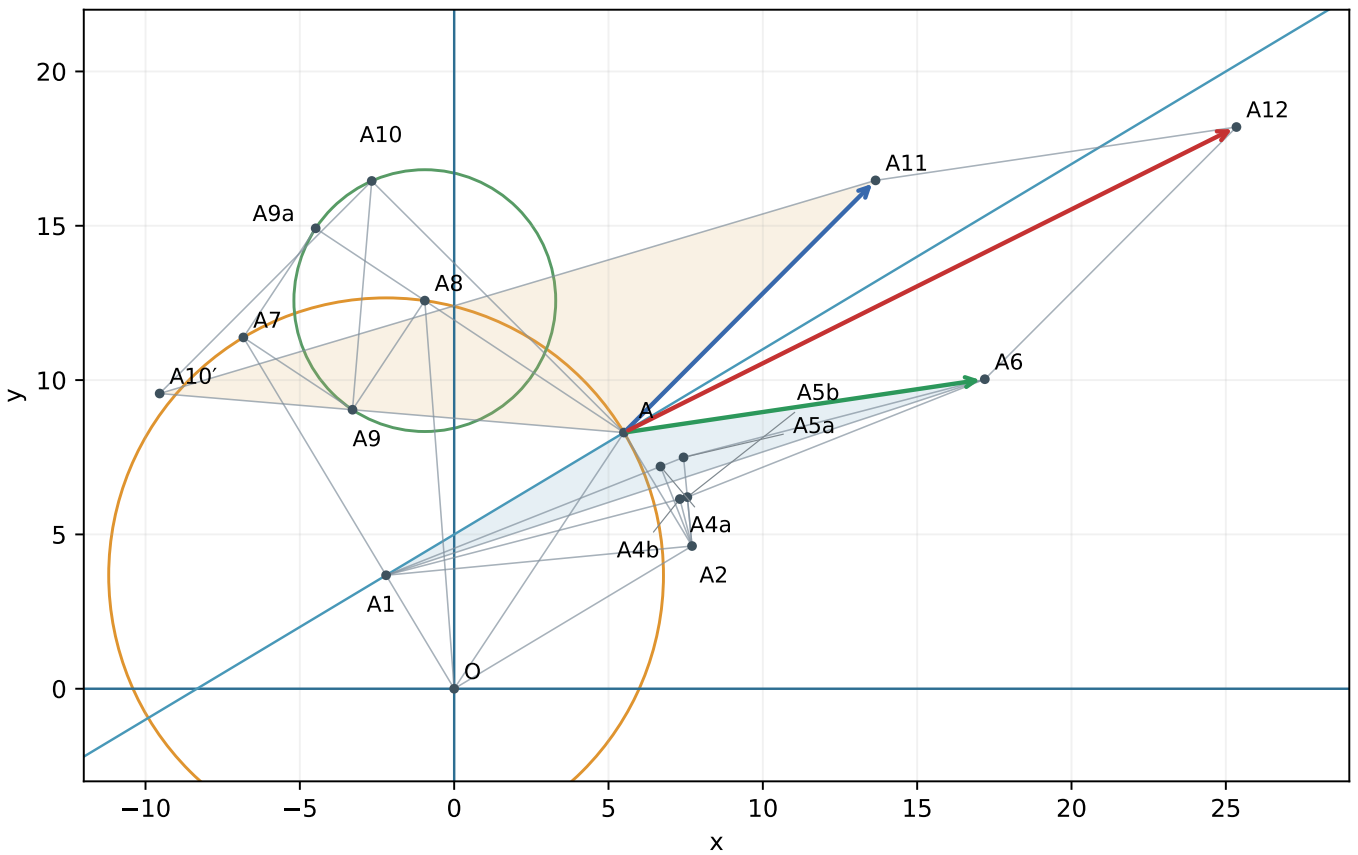
Source: Sangbunhak Z1 (formal spec by Kim Sangbok) + ChatGPT session computation | $a=3/5$, $b=6$, $x=3.5$

Z1 Sangbun Vector ($a=-5/6$, $b=-5$, $x=6.5$)

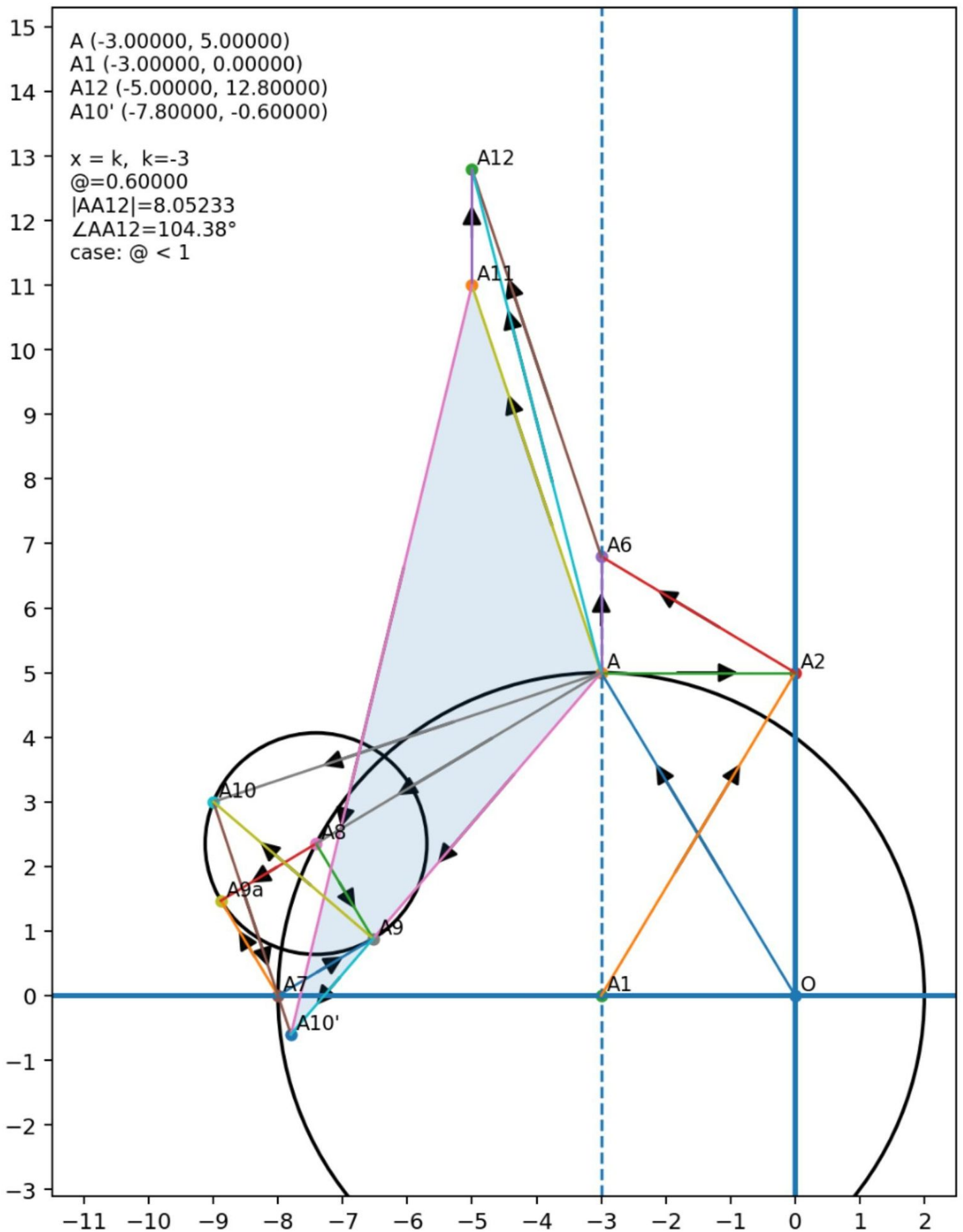


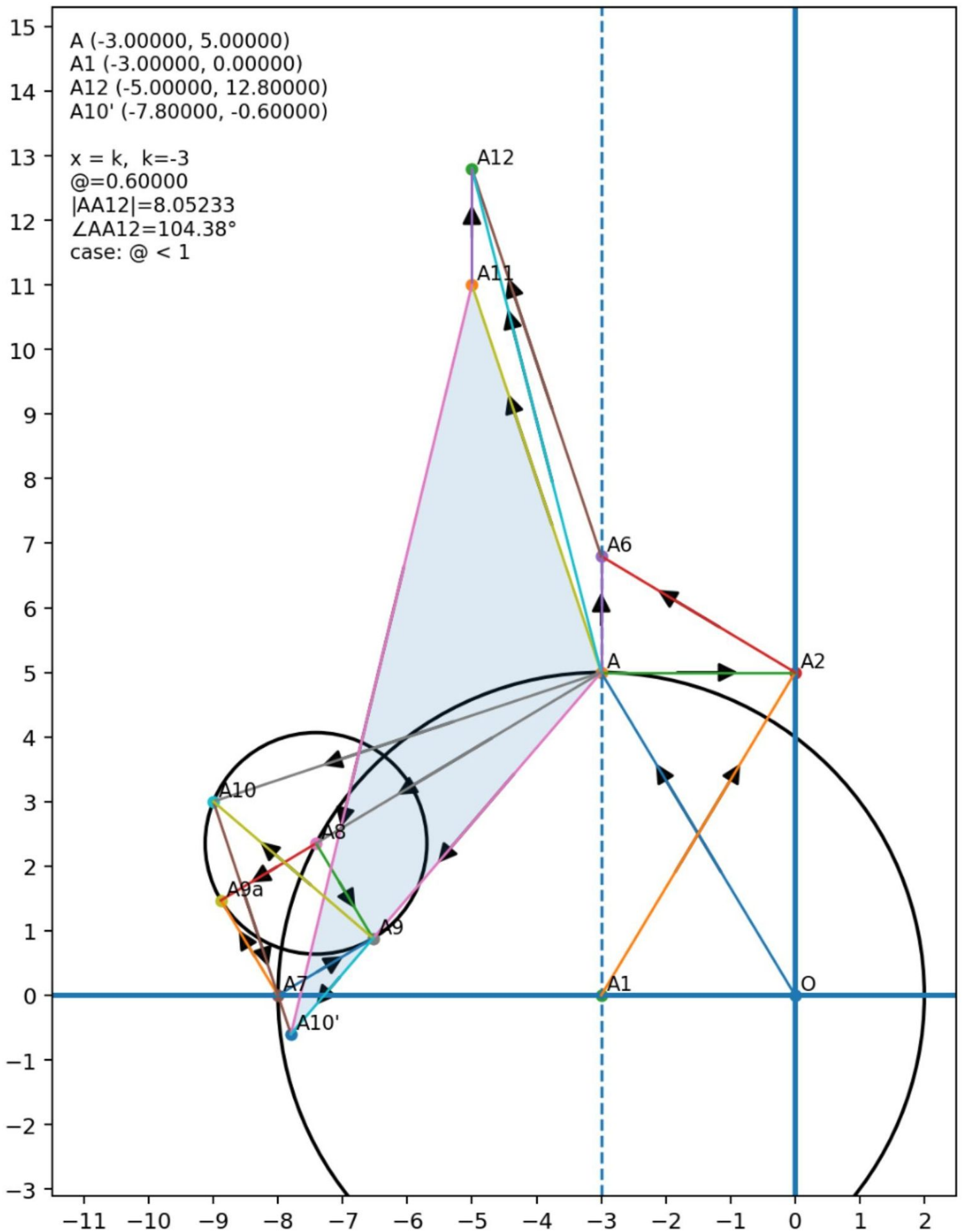
Z1 Sangbun Vector: Verified Example

$a = 3/5$, $b = 5$, $x = 5.5$, $@ = 0.47710 (< 1)$
 $A = (5.50000, 8.30000)$ $A1 = (-2.20588, 3.67647)$
 $A10' = (-9.54178, 9.56514)$
 $A12 = (25.34368, 18.19921)$
 $|AA12| = 22.17580$ $|A1A6| = 20.41115$
 $|A10'A11| = 24.19993$ $\angle AA12 = 26.51275^\circ$

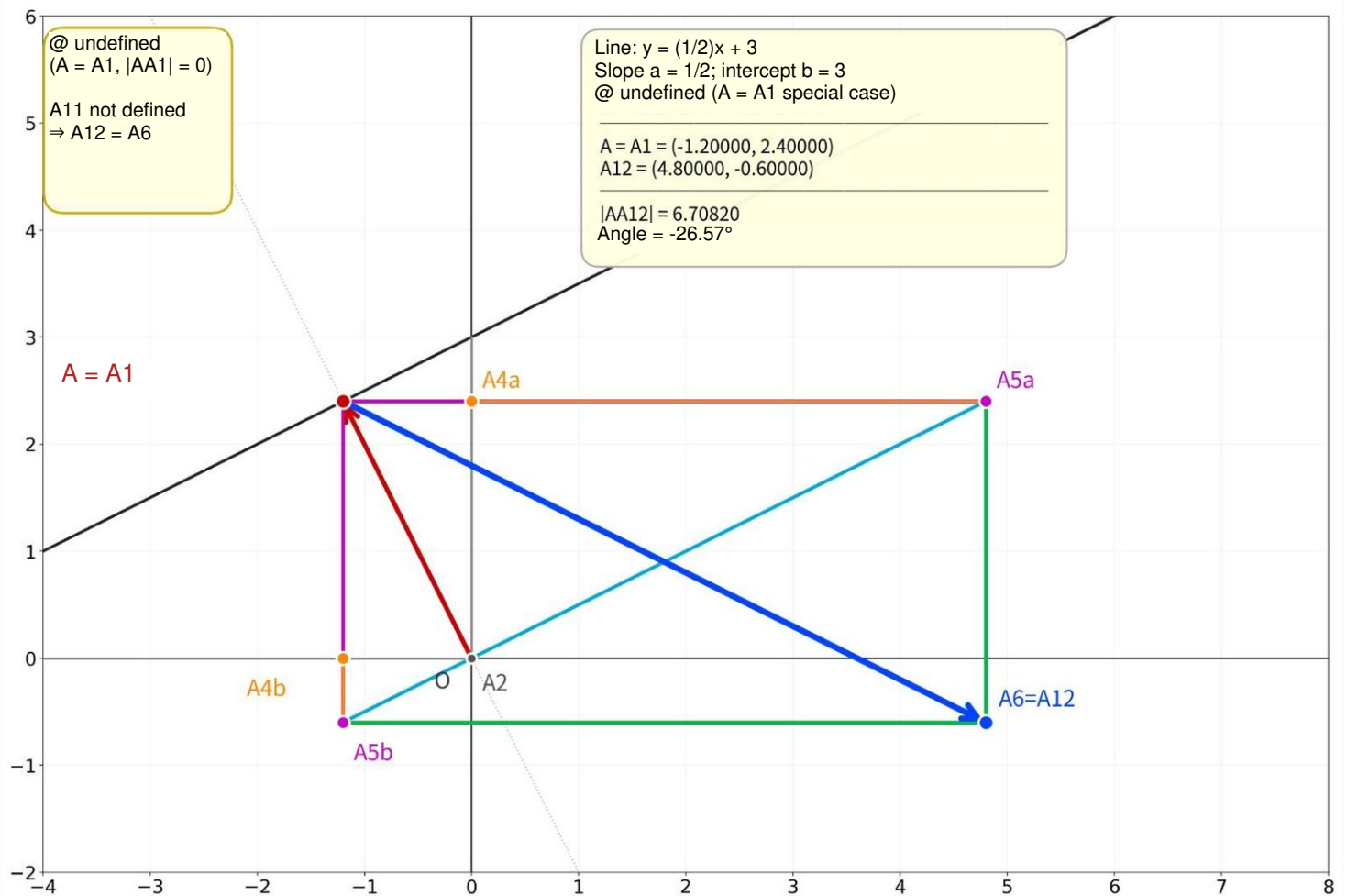


Green: AA6 · Blue: AA11 · Red: AA12
 Area(AA6A1) = 20.36830 · Area(AA11A10') = 66.60705





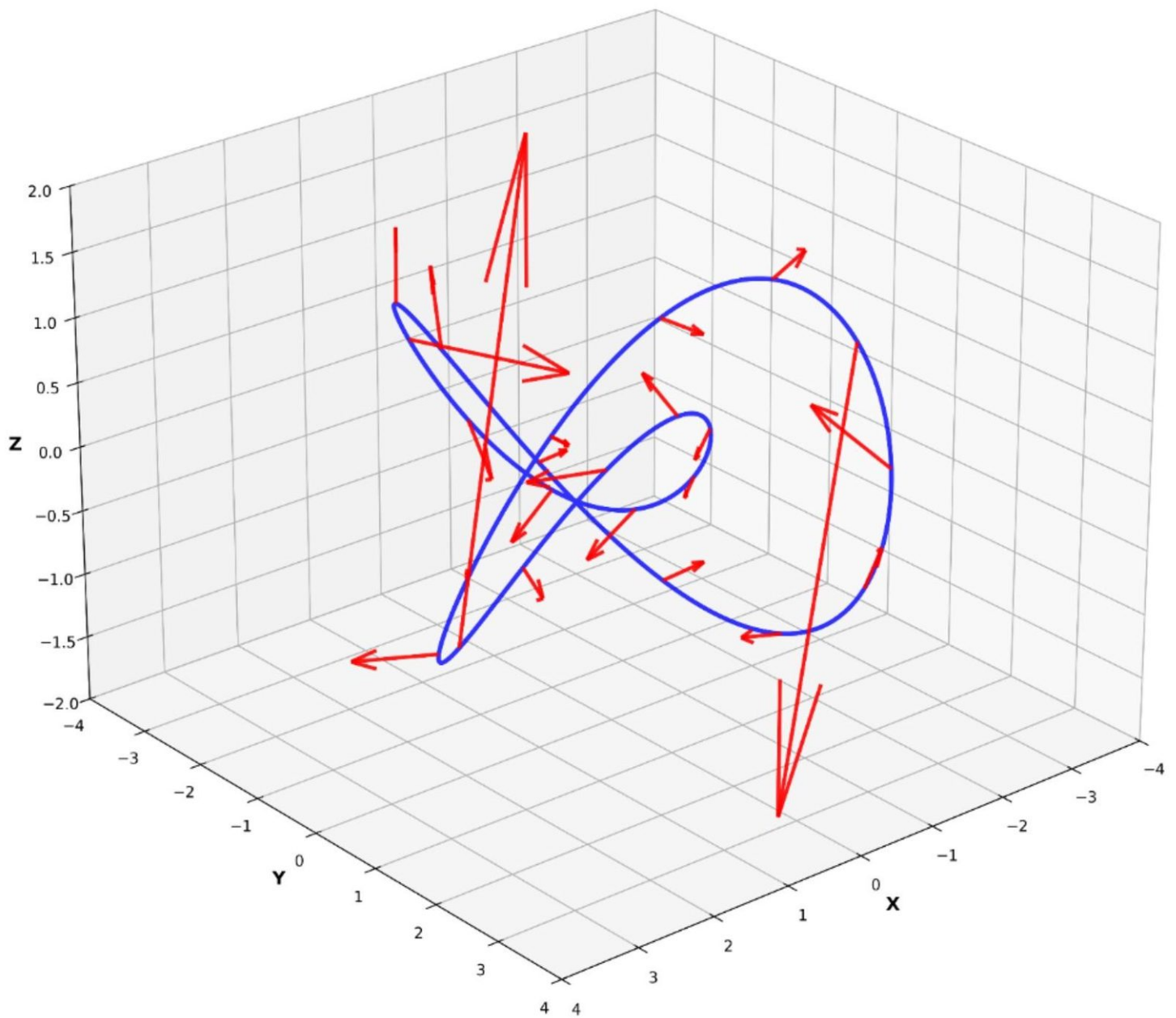
Sangbun Z1: $A = A1$ (special case)



In the special case $A = A1$, neither $A10$ nor $A11$ is constructed.
The prescribed rule gives $A12 = A6$.

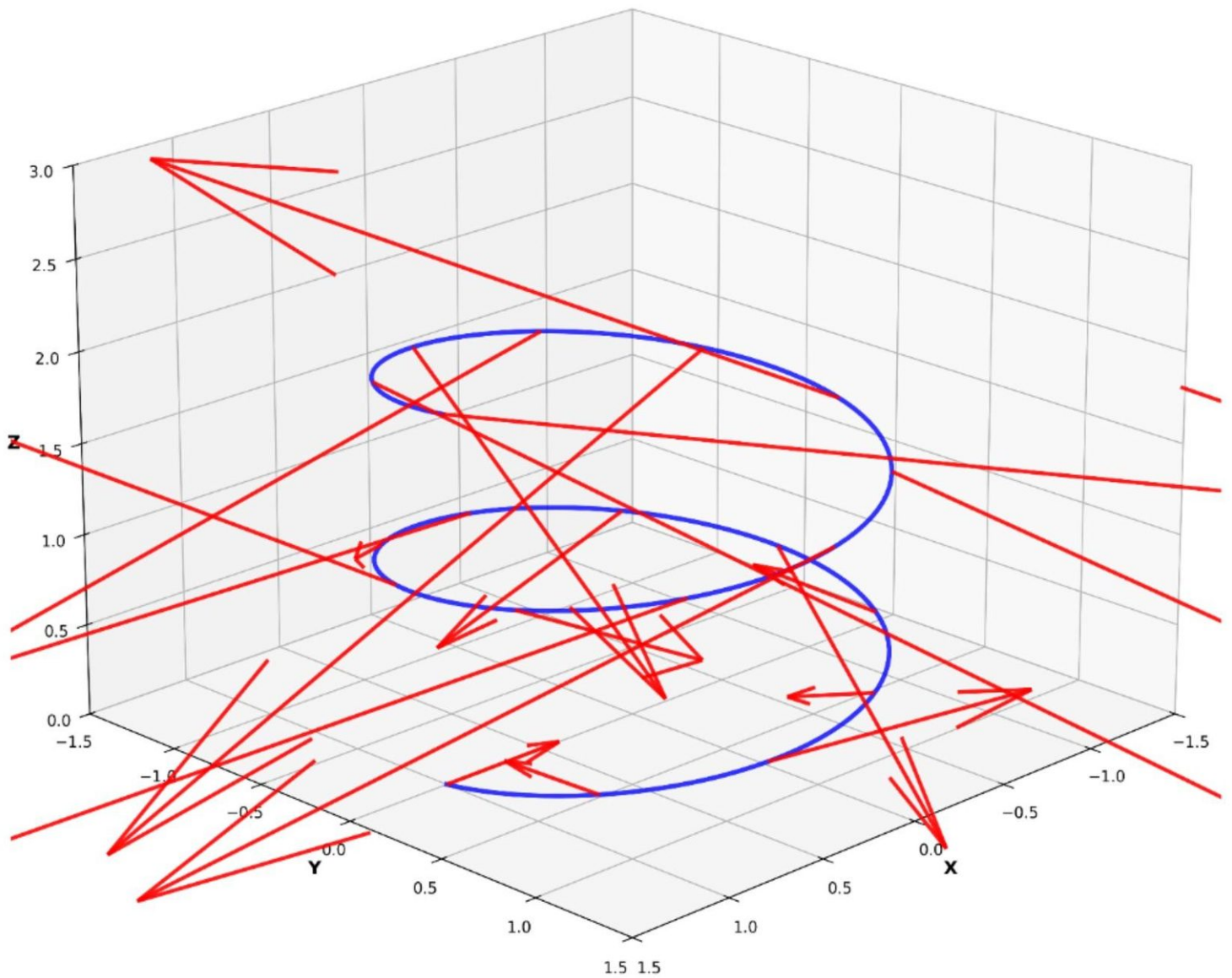
Trefoil Knot - FULL Z1 Sangbun Vector Field
(xy + yz + zx planes synthesis - Complete A1~A12 Algorithm)
Length: Avg=6.613, Max=24.995, Min=2.000, Std=6.132

— Curve



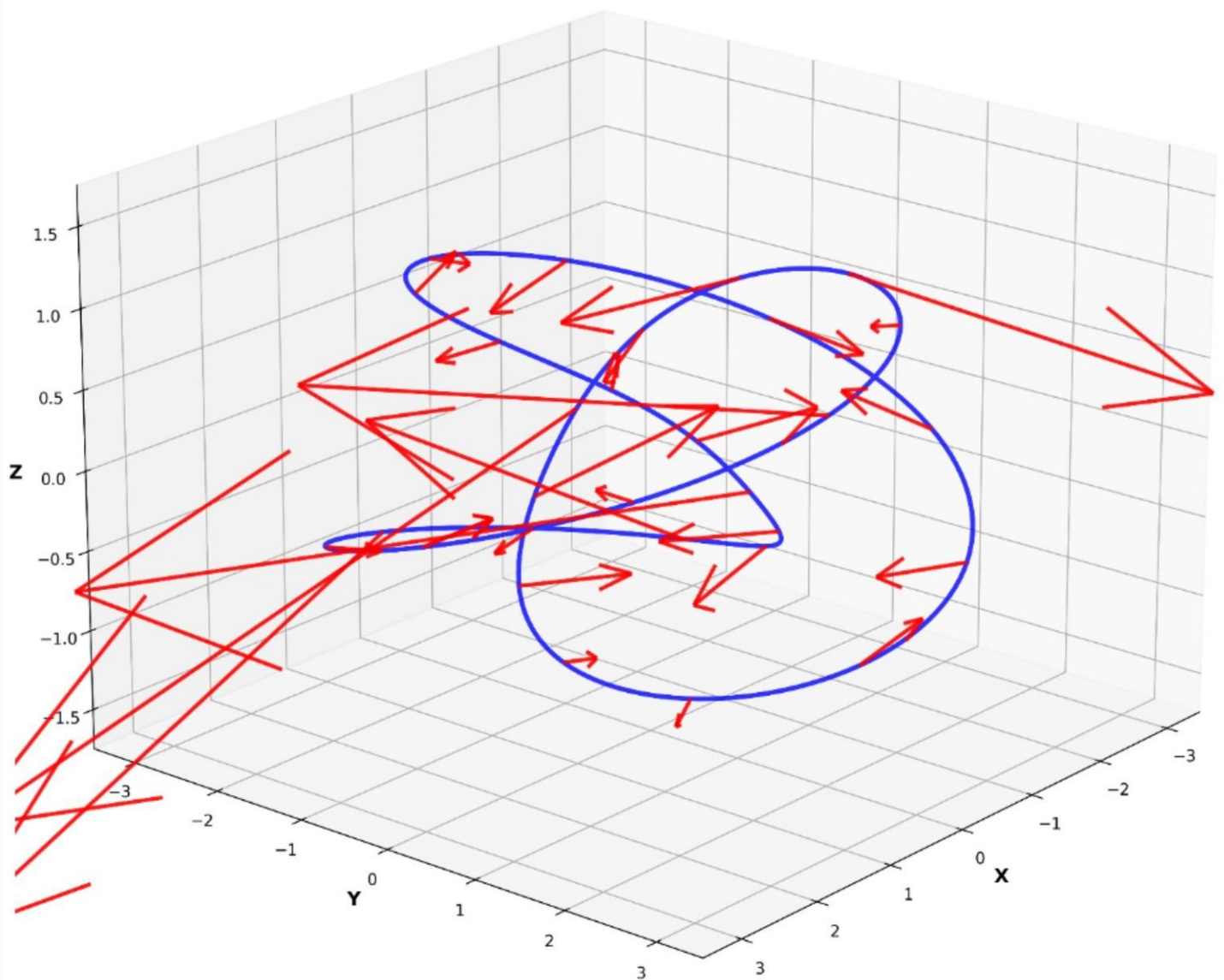
Helix - FULL Z1 Sangbun Vector Field
(xy + yz + zx planes synthesis - Complete A1~A12 Algorithm)
Length: Avg=48.284, Max=738.063, Min=1.572, Std=158.618

— Curve



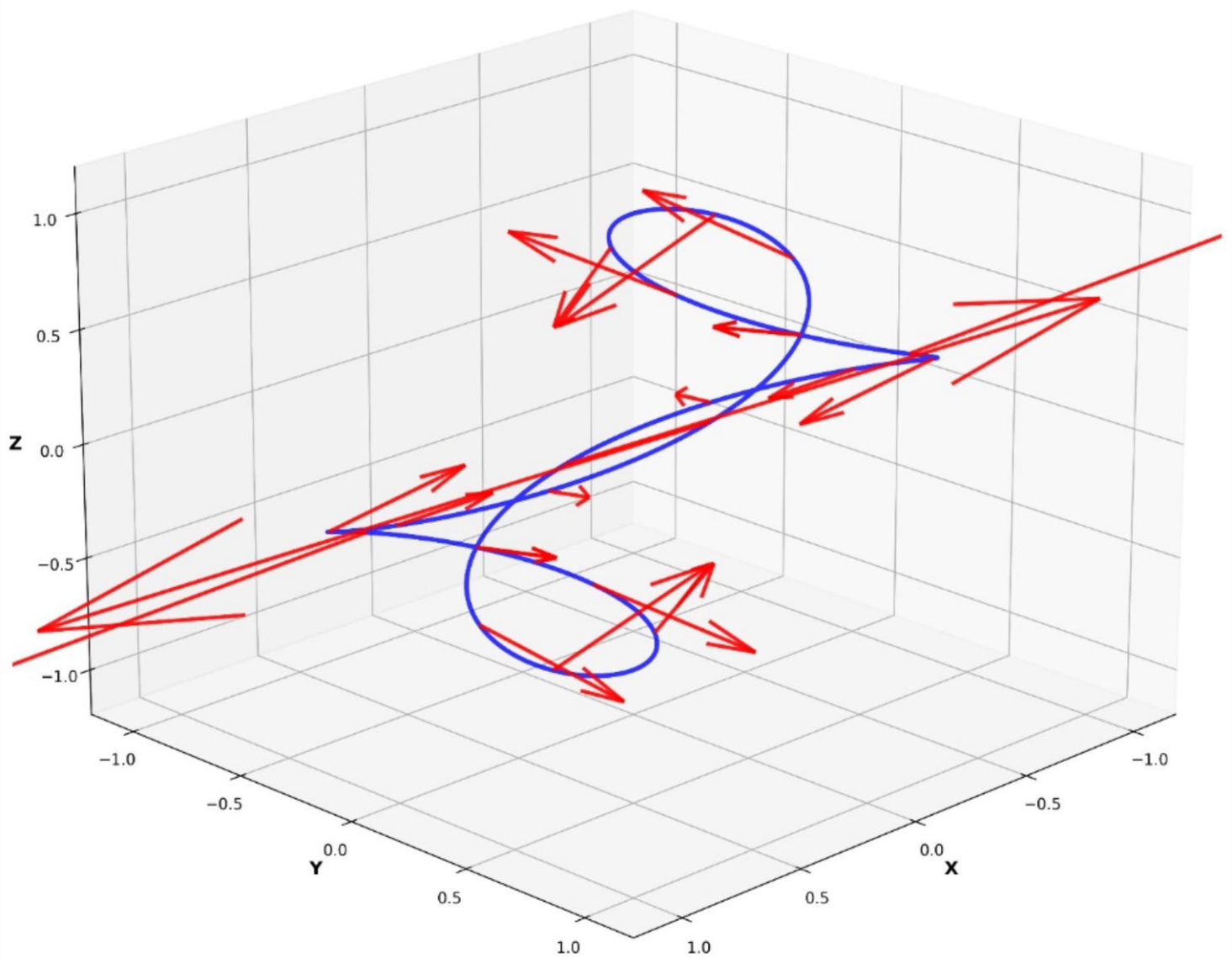
Torus Knot (3,2) - FULL Z1 Sangbun Vector Field
(xy + yz + zx planes synthesis - Complete A1~A12 Algorithm)
Length: Avg=13.481, Max=53.528, Min=3.082, Std=14.580

— Curve

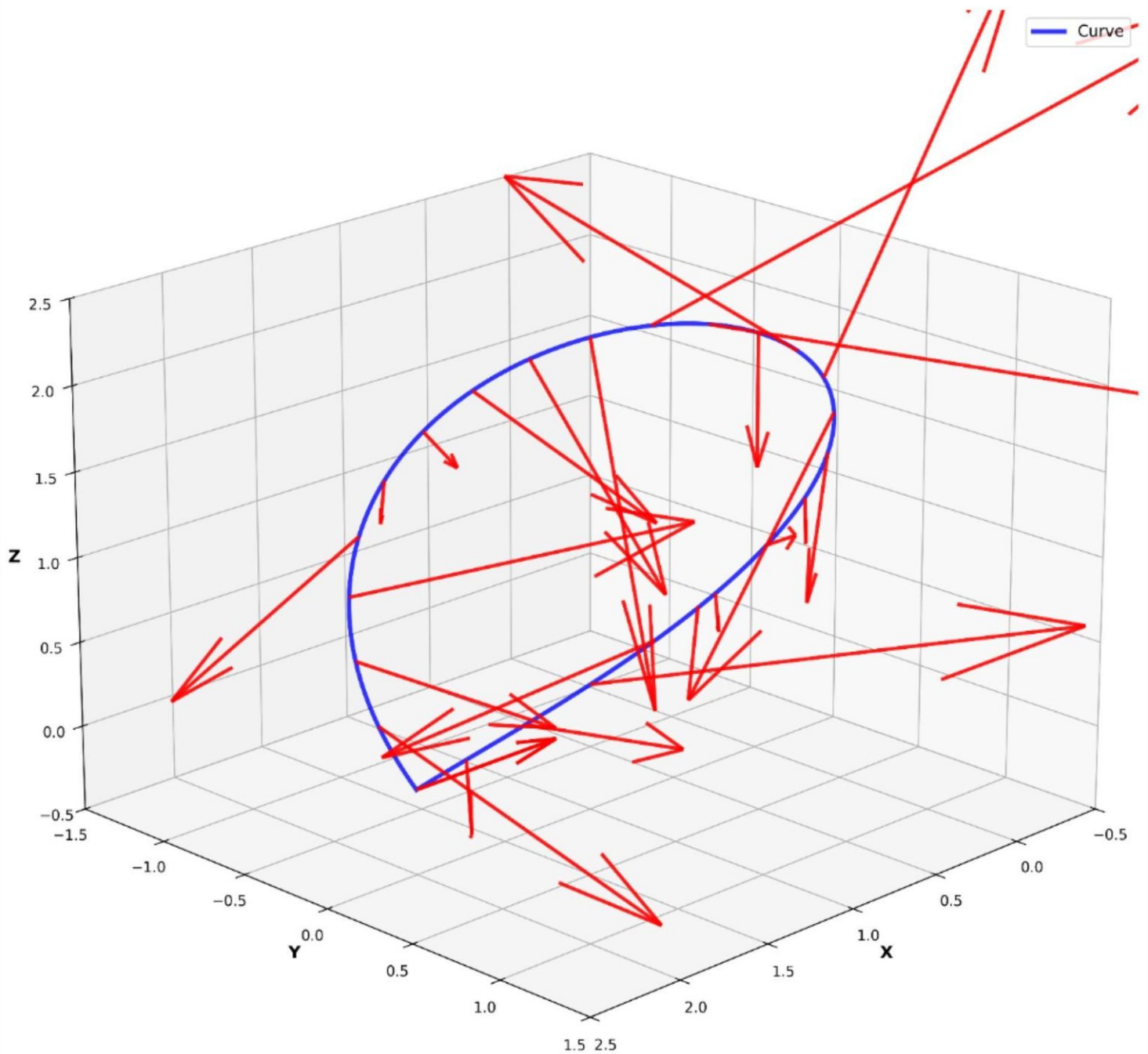


Spherical Spiral - FULL Z1 Sangbun Vector Field
(xy + yz + zx planes synthesis - Complete A1~A12)
Avg=3.990, Max=12.331, Min=0.000, Std=3.807

— Curve



Viviani's Curve - FULL Z1 Sangbun Vector Field
 (xy + yz + zx planes synthesis)
 Avg=7.908, Max=16.872, Min=1.659, Std=4.699



Cylindrical Helix (Elliptical) - FULL Z1 Sangbun Vector Field
(xy + yz + zx planes synthesis)
Avg=11.464, Max=31.773, Min=2.879, Std=6.831

— Curve

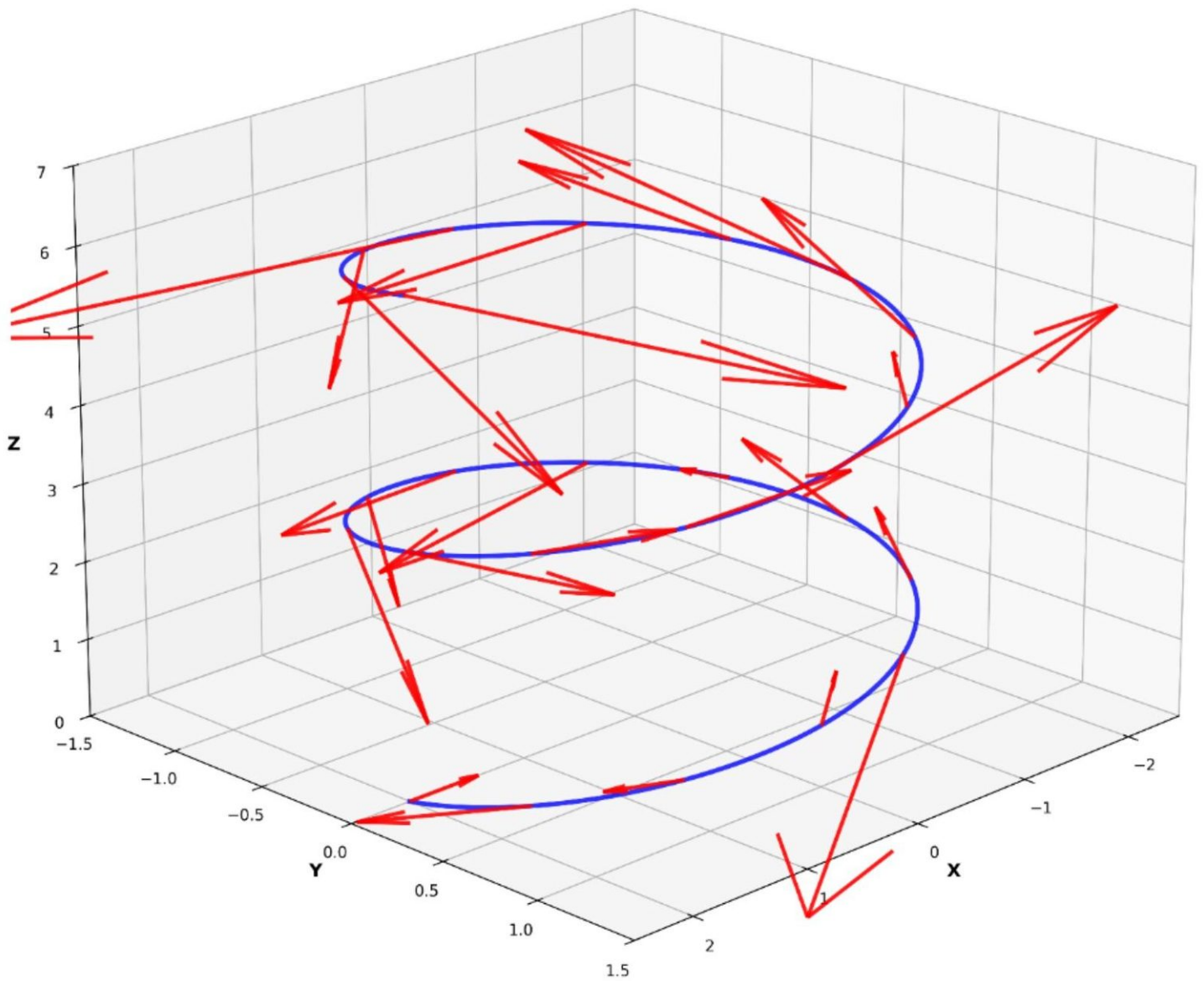
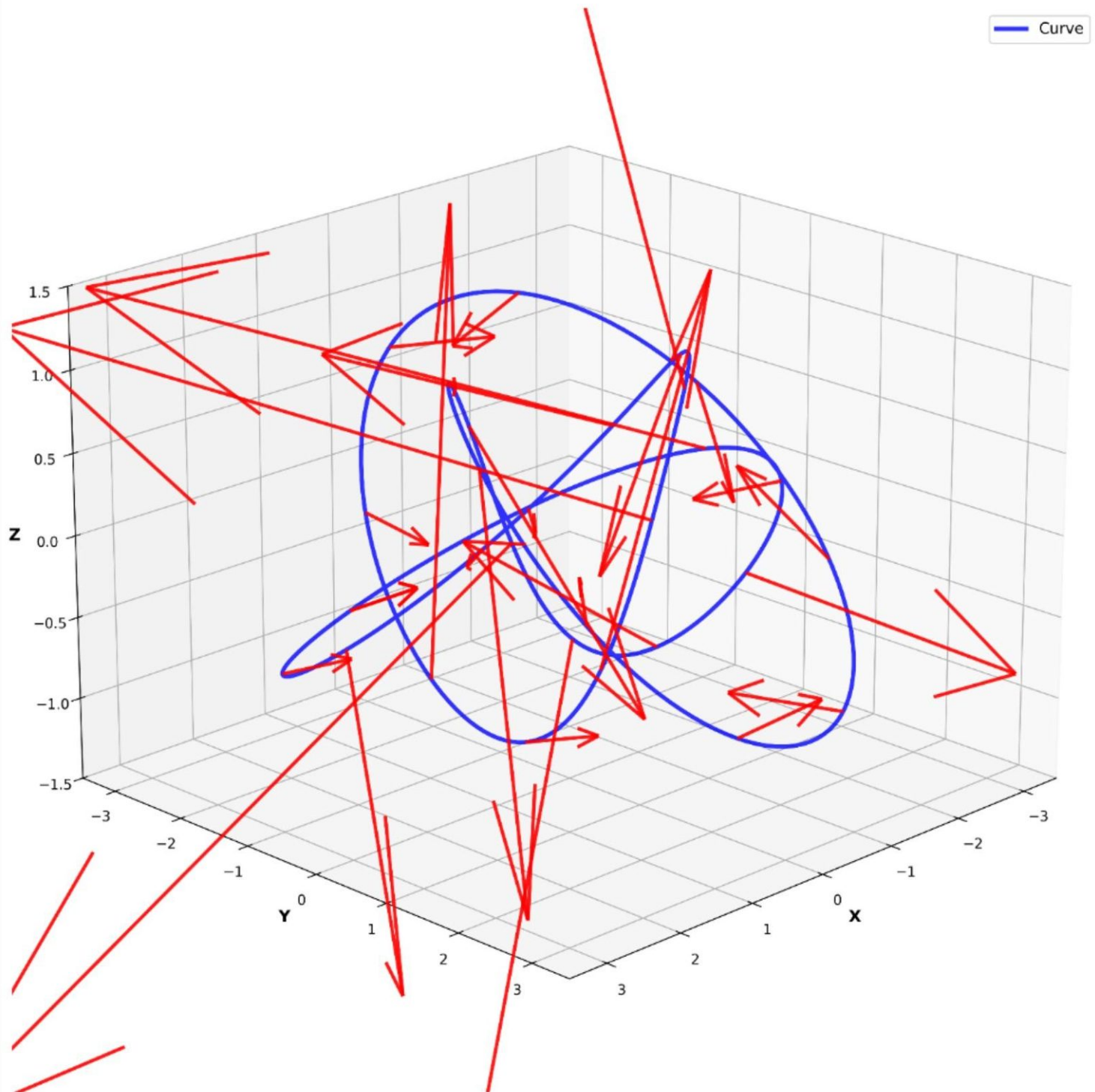
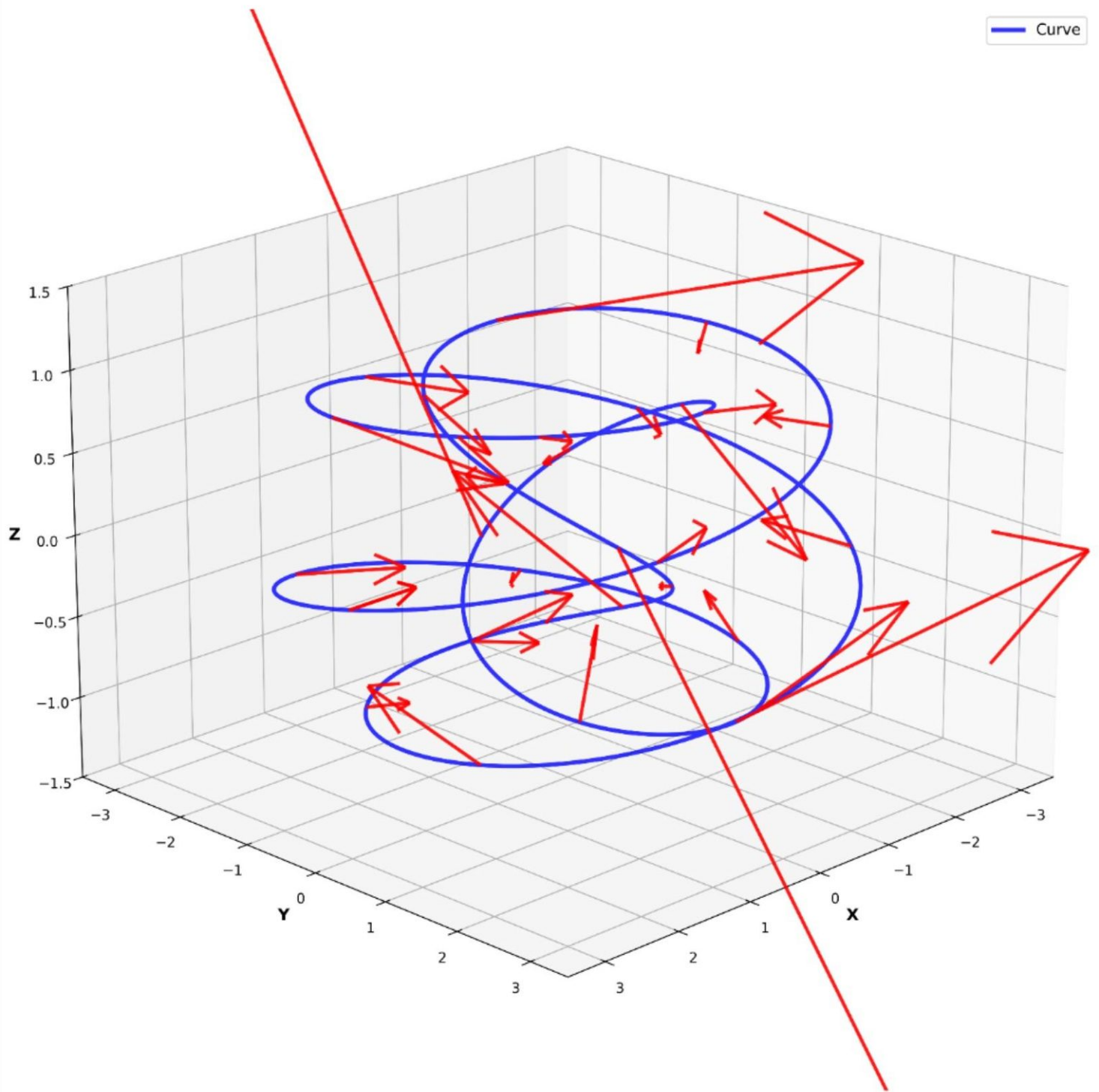


Figure-8 Knot - FULL Z1 Sangbun Vector Field
 (xy + yz + zx planes synthesis)
 Avg=17.275, Max=55.083, Min=3.644, Std=14.582



Cinquefoil Knot (5,2) - FULL Z1 Sangbun Vector Field
(xy + yz + zx planes synthesis)
Avg=17.566, Max=133.092, Min=2.062, Std=31.914



Lissajous Curve 3D - FULL Z1 Sangbun Vector Field
 (xy + yz + zx planes synthesis)
 Avg=15.001, Max=62.077, Min=0.000, Std=20.516

